From Form to Program:

Medium-Range Conditional Codon Syntax Inside Real Genes

Strict Null-Model Evidence Across 54 Cellular RefSeq Taxa,
a Codon-Mirror-Symmetry Control, Two-Scale Phylogenetic Transfer,
Intron–Exon Compartment Confinement, Synthetic Genome Forensics,
and the Biotech "Max-CAI" Codon-Optimization Crash

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Abstract

Statistical descriptions of coding DNA traditionally focus on macroscopic marginal invariants—nucleotide composition, isochore stratification, marginal codon usage, and single-stranded parity rules—which together define the static compositional profile (*form*) of a genome. This paper operationalizes the distinction between static *form* and dynamic *substance* in the language of held-out Markovian information theory and asks a strictly dynamical question: beyond marginal codon populations and strand-symmetry invariants, is there conditional sequential order inside real genes that behaves like a lineage-structured *program*? Using the self-testing `v14.5`/`v14.6` verification suite (`super_codon_autonomous_v14.py` and `super_codon_proof_v14_6.py`), we audit RefSeq coding sequences (CDS) across 54 unique cellular taxa (eukaryotic nuclear and prokaryotic RefSeq assemblies: a core panel of 13 model organisms plus an extended biosphere panel of 41 additional species; together, these 54 cellular taxa span mammals, non-mammalian vertebrates, insects, nematodes, land plants, green algae, fungi, protists, bacteria, cyanobacteria, and archaea) and one opt-in viral boundary control (SARS-CoV-2), using a local cache of 55 RefSeq assemblies analyzed across eight full audits. We enforce a true reading frame, one-gene-one-vote transcript deduplication, start/stop codon trimming, family-aware held-out gene splits, per-record alternate genetic code filtering (unit-tested across NCBI translation tables 1, 2, 3, 4, 5, 6, 11, and 12), and explicit small-sample guards. We evaluate the empirical codon-bigram transition model $P(c_{i+1} \mid c_i)$ on held-out genes against a ten-level hierarchy of null models: marginal codon usage (H0), boundary-nucleotide context (H1), amino-acid bigram syntax with marginal synonymous usage (H2), amino-acid syntax with empirical codon-pair usage (H2b), folded dinucleotide composition (H3), folded order-2 nucleotide trigram dynamics (H3b), block-shuffle local order at scales $L \in \{3, 10, 30, 100\}$ codons (H4), amino-acid-preserving H2b surrogates (H5), gene-specific synonymous multiset shuffles (H6), and a **mirror-pair shuffle** (H7) that preserves the exact per-gene populations of all 32 codon $\leftrightarrow$ reverse-complement mirror pairs (extended Chargaff's second parity rule at the codon level) while destroying sequential order. Across the sampled biosphere (spanning extreme compositions from GC3 = 0.1416 in *Dictyostelium discoideum* to GC3 = 0.9177 in *Thermus thermophilus*), all 52 cellular assemblies meeting the within-species sample-size guard (both training and held-out test splits containing ≥ 20 units and $\geq 50{,}000$ transitions) pass the full strict stack at $L = 30$ codons with positive 95% bootstrap confidence intervals on all seven strict-stack quantities (d_{2b}, d_3, d_{3b}, $\Delta_{H4}(30)$, Δ_5, Δ_6, and Δ_7), including the newly bootstrapped H6 and H7 controls (the two remaining small-split cellular assemblies likewise exhibit strictly positive point estimates and 95% bootstrap lower bounds across all seven criteria, with formal strict claims withheld solely because their test splits fall below the 50,000-transition guard). Cross-species transfer resolves into *two distinct phylogenetic scales*. At the *coarse* scale, a symmetric vertebrate-wide conditional layer ($d_{2b}^{\text{transfer}} > 0.02$ plus H4

gate) holds across 20/20 directed pairs in the 5-taxon core vertebrate bridge, 20/20 directed pairs in a five-taxon bridge spanning four vertebrate classes (mammals, birds, amphibians, and two teleost orders), and 132/132 directed pairs across 12 mammalian species from six orders. Outside vertebrates, this coarse threshold is reached mainly within narrow lower-rank clades (e.g. *Caenorhabditis*, core eudicots, monocot grasses, Enterobacteriaceae), with two exceptional high-GC prokaryotic transfers into *Pseudomonas aeruginosa* (*H. salinarum* and *M. tuberculosis*) and one coarse high-GC plant-to-alga transfer (*Zea mays* to *Chlamydomonas reinhardtii*). At the *fine* scale, codon-choice conservation measured by a target-scored H5 surrogate metric (gap $= \Delta_5{}^{\text{transfer}}_{\text{target}} - \Delta_5{}^{\text{within}}_{\text{target}}$, ratio $= \Delta_5{}^{\text{transfer}}/\Delta_5{}^{\text{within}}$) on a pre-specified focal set of 34 mammalian contrasts reveals a reproducible **Primate–Rodent directional asymmetry**: within the dedicated 12-mammal audit, Human is bidirectionally H5-conserved with Chimp, Gorilla, and Macaque; Mouse and Rat are bidirectionally conserved with each other; and Cow and Pig are bidirectionally conserved. Examined focal primate-to-rodent transfers exhibit small negative gaps (gap $\in [-0.0065, -0.0032]$, straddling the strict 0.005 and near-miss 0.012 tiers across runs), whereas every examined rodent-to-primate transfer exhibits a large positive overshoot (gap $\in [+0.0171, +0.0207]$) despite very small GC3 distance (ΔGC3 ≤ 0.011 across the core and focal mammal contrasts; ≤ 0.008 within the 12-mammal audit). Finally, an autonomous compartment-level control on fourteen human RefSeqGene loci (`v14.6`: the complete *p*53 tumor-suppressor family *TP53*, *TP63*, *TP73*; the complete *RAS* proto-oncogene family *KRAS*, *NRAS*, *HRAS*; and *BRCA1*, *PTEN*, *RB1*, *GAPDH*, *ACTB*, *HBB*, *MYC*, and *APOE*) demonstrates that within the exact same genomic locus, the conditional syntax residual d_{3b} is strictly positive in mature concatenated exons (panel mean +0.1721 nats/codon) and strictly negative in both excised introns (mean −0.3097 nats/codon; mean exon-minus-intron contrast +0.48182 nats/codon in **14/14** genes, $p = 2^{-14} = 6.1 \times 10^{-5}$) and whole unspliced pre-mRNA across all three reading frames (mean exon-minus-best-3-frame contrast +0.41617 nats/codon in **14/14** genes), while 32-pair mirror asymmetry (Mirror_{L_1}) is higher in spliced exons than in introns in **14/14** loci (mean 0.4176 vs. 0.1615). Within spliced *TP53*, exon-by-exon profiling shows that the mutable DNA-binding core (exons 5–8, codons 125–300, where 86% of cancer mutations cluster) carries a 2.34× higher d_{2b} (+0.0989 vs. +0.0423) and 1.78× higher d_{3b} (+0.1454 vs. +0.0816) than flanking exons, while in all three 570-nt *RAS* paralogs positive d_{3b} is concentrated in the catalytic GTPase exons 1–3 (9/9 exons positive) and single-codon oncogenic hotspot substitutions (*TP53* `R175H`, `G245S/D`, `R248Q/W`, `R273H/C`, `R282W`; *KRAS*/*NRAS*/*HRAS* codons 12, 13, and 61) resolve into either direct syntactic program collapse ($\Delta d_{3b} < 0$, reaching −1.2117 nats at `R282W` and flipping sign at `R175H` and *KRAS* `G12C`) or CpG/tandem-glycine mutational-trap relaxation ($\Delta\text{LL}_{\text{emp}} > 0$). These results support an operational compression-operator interpretation of splicing: intronic and unspliced compartments act as relatively higher-entropy, more mirror-symmetric compositional buffers (*form*), whereas exon concatenation enriches a symmetry-broken, more time-directed coding *program* (with the intronic baseline and sequence-level geometric invariants completely independent of CDS training data, and single-locus overlap representing $< 10^{-3}$ of the pooled reference transitions). Finally, an out-of-sample frontier audit (`super_codon_proof_v14_6_7.py`, `v14.7`) across 13 boundary genomes demonstrates that (i) all four examined giant DNA viruses (*Mimivirus*, *Megavirus chilensis*, *Pandoravirus salinus*, and 30,000-year permafrost *Pithovirus sibericum*), all four deep subsurface and abyssal hyperthermophiles (including *Candidatus Desulforudis audaxviator* at 2.8 km depth and *Methanopyrus kandleri* at 122°C), and the Asgard archaeon *Candidatus Prometheoarchaeum syntrophicum* MK-D1 pass all 7/7 strict bootstrap criteria, whereas the 160-kb minimal endosymbiont *Candidatus Carsonella ruddii* (GC3 = 0.0757) loses synonymous d_{2b} and Δ_5 under extreme reductive evolution; and (ii) scoring Venter's synthetic genomes (*JCVI-syn1.0* and *JCVI-syn3.0*, translation table 4) under the wild-type *Mycoplasma mycoides* model confirms 100% program retention across all 412 essential genes in *JCVI-syn3.0* ($d_{3b} > 0$), while blindly pinpointing the human-inserted foreign vector markers in *JCVI-syn1.0* (*E. coli* `lacZ` $d_{3b} = -0.4564$, yeast `HIS3` −0.4286, and `bla` β-lactamase −0.2353) and synthetic intergenic watermark cassettes ($d_{3b} \in [-0.4825, -0.1964]$ across all reading frames). Finally, a four-module translational and evolutionary forensics audit (`super_codon_proof_v14_8.py`, `v14.8`) establishes that: (i) standard industrial "codon optimization" (`Max-CAI`, which replaces every sense codon with the single most frequent human synonymous codon while preserving 100% of the amino-acid sequence) crashes the conditional codon-pair syntax residual d_{2b} from a wild-type mean of +0.0427 to strictly negative values across **14/14** benchmark human genes

(panel mean -0.0589 nats/codon, $p = 2^{-14} = 6.1 \times 10^{-5}$; e.g. *GAPDH* $+0.1168 \to -0.0904$, *HBB* $+0.0814 \to -0.1138$, *TP53* $+0.0677 \to -0.0340$) and collapses 64-codon Shannon entropy from 5.322 to 4.091 bits, whereas first-order Markov compilation (`Markov_H2b`) preserves positive program syntax ($+0.1129$ vs. $+0.0684$ for memoryless `H0` synonymous sampling); (ii) human DNA viruses (Epstein–Barr virus, Adenovirus C2, HPV-16, Vaccinia) exhibit strictly positive d_{3b} on the human transition matrix ($+0.0970$ to $+0.1145$) and negative d_{3b} on *E. coli* (-0.0377 to -0.0952), with blind per-gene scoring isolating specific host-mimicking "Trojan" effectors—most notably the transforming oncoprotein *E7* as the sole gene in HPV-16 with positive human synonymous syntax ($d_{2b} = +0.0327$), the Adenovirus immune-evasion stealth protein *E3 RID-α* ($d_{2b} = +0.0958$, 1.94× the native human mean), and EBV-pirated nucleotide enzymes *BLLF3* ($+0.0538$) and *BaRF1* ($+0.0517$); and (iii) while 93.1% (d_{2b}) and 94.4% (d_{3b}) of 1,500 scanned human genes are best-scored by vertebrate matrices (0/1,500 non-vertebrate outliers), endosymbiotic and retroviral acquisitions follow a quantitative assimilation clock—with **65/67** (*Arabidopsis* chloroplast, 97.0%, $p = 3.9 \times 10^{-17}$) and **10/13** (human mitochondrial) organellar genes rewritten from bacterial ancestor syntax to host nuclear syntax, ancient imports (*RAG1*, *CS*) fully assimilated ($d_{2b} \geq +0.0724$), and the younger primate placental retroviral envelope *ERVW-1/Syncytin-1* at an intermediate stage ($d_{2b} = +0.0288, d_{3b} = +0.0304$). Extreme compositional contrasts (such as the haloarchaeon *H. salinarum*, GC3 = 0.8630) are cleanly isolated by a paired GC3/acidic-proteome distance control as compositional hardware mismatches rather than shared syntax.

Keywords: codon usage bias, conditional entropy, Markov null models, block shuffle, synonymous shuffle, mirror-pair shuffle, Chargaff parity rule, CDS true frame, alternate genetic codes, RefSeq, RefSeqGene, information theory, two-scale phylogenetic transfer, directional asymmetry, GC3 artifact control, intron–exon confinement, *TP53* splicing trajectory, splice-junction phase, compression operator, comparative genomics.

1 Introduction

Macroscopic genome statistics are necessary but insufficient to characterize the sequential organization of protein-coding DNA [1, 2]. A coding sequence can satisfy stable global nucleotide frequencies, isochore GC equilibria, marginal codon usage biases, dinucleotide preferences, and amino-acid composition while lacking any higher-order conditional dependence between successive codons beyond those marginals. Conversely, apparent sequential correlations in raw genomic windows often arise as artifacts of reading-frame mixture, isoform duplication, translation initiation/termination signals, or local nucleotide trigram inertia. The rigorous information-theoretic question [1] is therefore: *inside true-frame coding sequences, does the choice of the next codon c_{i+1} depend on the preceding codon c_i in a way that survives simultaneous controls for marginal composition, order-2 nucleotide dynamics, amino-acid sequence, local codon-pair bias, gene-specific synonymous usage, and strand-parity symmetry?*

Conceptualizing this problem benefits from distinguishing two complementary layers of genomic organization: a static profile (*form*) and a dynamical sequential profile (*substance* or *program*). In the literature on whole-genome codon statistics and numerical genomics, previous studies by Perez and collaborators [4, 5, 6, 7, 8, 9, 10, 11, 12, 13] analyzed marginal codon populations, atomic-mass strand equilibria, and long-range Fibonacci/Lucas and Master Code resonances in DNA (systematized in *Codex Biogenesis: Les 13 codes de l'ADN* [4]), documenting that the 64 codons partition into 32 reverse-complement mirror pairs obeying an extended Chargaff second parity rule at the codon level (with odd- and even-ranked mirror partners balanced at 474,337,193 vs. 473,466,674 triplets, $r = 0.999995$, across the whole human genome [6, 8]), demonstrating $3n$-nucleotide periodic codon emergence across spliced *BRCA1* transcripts [8], mapping codon-level mutability hotspots across the 393 sense codons of human *TP53* (specifically isolating the exons 5–8 DNA-binding core and hotspot codons 175, 245, 248, and 273) [9], demonstrating that chemical genome synthesis and transposon deletion in artificial *Caulobacter ethensis* (*C. eth-2.0*), Ven-

ter's *Mycoplasma mycoides* constructs (*JCVI-syn1.0*, *JCVI-syn3.0*, *JCVI-syn3A*), recoded *E. coli* (*Syn61*), and synthetic yeast Chromosome XII (*Sc2.0 synXII*) disrupt whole-genome palindromic Fibonacci and Lucas resonance equilibria [11], contrasting the 570-nucleotide coding sequences of the three human *RAS* proto-oncogenes (*KRAS*, *NRAS*, *HRAS*) and their sensitivity to single-nucleotide oncogenic mutations [13], and formalizing the static combinatorial compression of 64 single-strand (128 double-strand) codons into 21 functional channels (20 amino acids plus stop; Theorem 8 in [12]); Petoukhov and Svirin [14] (see also the joint Mannheim CAMMBIO symposium [8]) developed algebraic genomatrix representations validated on those marginal codon counts; and Bernardi [15] established the compositional isochore framework governing regional GC stratification. All three lines of inquiry describe *statics*: they characterize marginal counts, compositional equilibria, and pairwise parity symmetries of codon populations. They do not measure sequential transition order.

The present work translates the form/substance distinction into testable, held-out Markovian dynamics on curated NCBI RefSeq coding sequences [3]. We treat compositional marginals, nucleotide trigram statistics, and reverse-complement mirror-pair populations as the *static form* of a coding sequence, and we design a strict hierarchy of null models to absorb them completely. The central object of our audit is the smoothed empirical conditional distribution

$$P(c_{i+1} \mid c_i),$$

trained exclusively on 80% of a species' deduplicated genes and evaluated on the held-out 20% test genes. We test whether real coding sequences carry excess conditional predictability (in nats per codon transition) over ten increasingly restrictive baselines:

1. **H0:** marginal codon frequencies;
2. **H1:** codon boundary nucleotide context $P(c_{i+1} \mid n_3(c_i))$;
3. **H2:** amino-acid bigram syntax with marginal synonymous usage;
4. **H2b:** amino-acid bigram syntax with empirical codon-pair usage [16, 17];
5. **H3:** folded dinucleotide composition;
6. **H3b:** folded nucleotide trigram (order-2 Markov) sequential dynamics;
7. **H4:** intra-gene block-shuffled sequences at block lengths $L \in \{3, 10, 30, 100\}$ codons;
8. **H5:** amino-acid-preserving surrogates generated from the training H2b conditional sampler;
9. **H6:** intra-gene synonymous shuffles preserving the exact amino-acid sequence and per-gene synonymous codon multiset;
10. **H7:** intra-gene mirror-pair shuffles preserving the exact count of each of the 32 codon $\leftrightarrow$ reverse-complement pairs of Perez [6] while destroying sequential order.

Using the hardened, self-testing `super_codon_autonomous_v14.py` (`v14.5`) pipeline, we execute this audit across 13 core reference organisms, a viral small-sample boundary control (SARS-CoV-2), and a curated 41-taxon extended biosphere panel (totaling 54 unique cellular taxa and 55 RefSeq assemblies). We show that medium-range conditional codon syntax at $L = 30$ codons is a ubiquitous within-species invariant of cellular coding sequences across the sampled biological GC3 spectrum ($0.1416 \leq \mathrm{GC3} \leq 0.9177$), and that cross-species transfer of this syntax is organized into a two-scale phylogenetic hierarchy: a broad, symmetric vertebrate macro-layer ($d_{2b} > 0.02$) and a fine, order-structured mammalian codon-choice core exhibiting a reproducible Primate–Rodent directional asymmetry.

Finally, to test whether this conditional syntax is strictly confined to the protein-coding compartment rather than reflecting local chromosomal background, we introduce an autonomous intron–exon and splice-junction audit in `super_codon_proof_v14_6.py` (`v14.6`, Section 6.5). By directly contrasting (1) the whole unspliced gene sequence (introns + exons across all three reading frames), (2) the mature coding sequence formed by concatenated exons, and (3) excised introns across canonical human RefSeqGene loci (*TP53*, *BRCA1*, *HBB*, and the 14-gene panel encompassing both the *p*53 and *RAS* paralog families), we show that introns and unspliced pre-mRNAs behave as relatively higher-entropy, more mirror-symmetric compositional compartments (*form*) with negative

d_{3b} across all reading frames, whereas concatenated coding exons exhibit strictly positive d_{3b} and stronger mirror-symmetry breaking in **14/14** loci, supporting an operational compression-operator transition from intronic *form* to exonic *program*.

2 Conceptual framework: static form versus conditional program

2.1 Marginal parity statics versus sequential transition dynamics

Two sequences of length N can share identical nucleotide composition, identical marginal frequencies of all 64 codons, and identical populations of all 32 reverse-complement codon pairs while differing completely in their transition matrix $P(c_{i+1} \mid c_i)$. Marginal, parity, and Master Code descriptions [4, 5, 6, 7, 8, 9, 10, 11, 12, 13] quantify zero-order (1-gram) properties of a sequence; conditional entropy and log-likelihood ratios [1] quantify first-order and medium-range sequential dependencies.

By operationalizing static properties as explicit null models, we prevent confounding between compositional equilibria and sequential syntax:

- **Compositional and sequential nucleotide form** is absorbed by H0 (codon marginals), H1 (boundary nucleotide), H3 (folded dinucleotides), and H3b (folded nucleotide trigrams) [15, 14].
- **Strand-parity form** of Perez [6] (Sec. 2.2 therein) is absorbed by H7, which permutes codons within each of the 32 unordered mirror pairs $\{c, \mathrm{revcomp}(c)\}$ inside every held-out gene.
- **Primary protein and local codon-pair constraints** are absorbed by H2, H2b, H4, H5, and H6.

A statistically significant positive residual across this entire stack demonstrates that coding sequences carry conditional transition information that cannot be reduced to static composition, strand symmetry, or amino-acid order.

2.2 Two scales of cross-species transfer

A critical methodological distinction introduced in our protocol is that cross-species syntactic conservation operates at two distinct resolutions:

1. **Coarse conditional transfer ($d_{2b}^{\mathrm{transfer}}$):** Does an empirical codon-bigram model trained on source species S predict held-out genes of target species T significantly better than S's amino-acid-plus-codon-pair null H2b, dinucleotide null H3, nucleotide-trigram null H3b, and $L = 30$ block-shuffle null H4? In particular, exceeding $d_{2b}^{\mathrm{transfer}} > 0.02$ (`coarse_clean_0p02`) measures whether two taxa share the macroscopic backbone of codon-pair conditional succession.
2. **Fine codon-choice conservation (target-scored H5 gap and ratio):** When synonymous choices along target T's held-out proteins are generated by source S's H2b conditional sampler and scored under target T's own empirical model ($\Delta_{5T}^{\mathrm{transfer}}$), how closely does that score match target T's native within-species surrogate baseline ($\Delta_{5T}^{\mathrm{within}}$)?

Separating these two scales resolves apparent paradoxes in comparative codon usage: vertebrates share a coarse conditional backbone across examined classes (teleost fishes, amphibians, birds, mammals), whereas fine synonymous choice is partitioned into order-level mammalian dialects with reproducible directional asymmetries.

2.3 Scope of the dynamical audit

This study is strictly scoped to sequential transition dynamics inside annotated coding sequences and their corresponding RefSeqGene exon–intron pre-mRNA spans. We do not test, parameterize, or make claims regarding whole-chromosome intergenic static ratios or numerical attractors proposed in whole-genome marginal studies [4, 5, 6, 7, 10, 12]; those concern zero-order marginal statics on different genomic objects and lie outside the scope of a conditional Markovian audit. Similarly, while our null stack and intron–exon compartment controls rule out mutational, reading-frame, and compositional artifacts, establishing adaptive natural selection requires orthogonal evolutionary rates (dN/dS on aligned orthologs) and functional translation assays.

3 Data and verification protocol (`v14.5` and `v14.6`)

3.1 Core and extended RefSeq coding-sequence panels and RefSeqGene loci

All multi-species analyses are performed on official NCBI RefSeq `*_cds_from_genomic.fna.gz` assemblies [3]. The primary audit panel comprises 13 cellular model organisms spanning mammals, non-mammalian vertebrates, invertebrates, plants, fungi, bacteria, and archaea, plus SARS-CoV-2 as an opt-in viral boundary control (Table 1). To test whether our findings generalize across the biosphere, we also curated and downloaded an extended panel of 41 additional cellular RefSeq assemblies (`--include-extended`), bringing the total local cache to 54 unique cellular taxa and 55 validated RefSeq archives (Section 6.4 and Appendix C). For the compartment-level intron–exon audit (Section 6.5), `super_codon_proof_v14_6.py` autonomously fetches and parses official human NCBI RefSeqGene records (`NG_017013.2` for *TP53*, `NG_005905.2` for *BRCA1*, `NG_000007.3` for *HBB*, and eleven additional benchmark human loci including the complete $p53$ and *RAS* paralog families) against the human reference transition model built from the `T2T-CHM13v2.0` CDS set (`GCF_009914755.1`, `max_records=6000`).

Species	Clade	RefSeq assembly directory	Role in audit
Human_T2T	mammal	`GCF_009914755.1_T2T-CHM13v2.0`	reference primate / mammal
Chimp	mammal	`GCF_002880755.1_Clint_PTRv2`	close hominid relative
Mouse	mammal	`GCF_000001635.27_GRCm39`	rodent outgroup (asymmetry test)
Chicken	vertebrate	`GCF_016699485.2_bGalGal1.mat...GRCg7b`	avian vertebrate bridge
Zebrafish	vertebrate	`GCF_000002035.6_GRCz11`	teleost vertebrate root
Drosophila	insect	`GCF_000001215.4_Release_6_plus...`	dipteran insect control
C. elegans	nematode	`GCF_000002985.6_WBcel235`	nematode isolation control
Arabidopsis	plant	`GCF_000001735.4_TAIR10.1`	eudicot plant control
Yeast	fungus	`GCF_000146045.2_R64`	fungal eukaryote control
E. coli	bacterium	`GCF_000005845.2_ASM584v2`	Gram-negative bacterial control
B. subtilis	bacterium	`GCF_000009045.1_ASM904v1`	Gram-positive bacterial control
M. jannaschii	archaeon	`GCF_000007745.1_ASM774v1`	methanogenic archaeon
H. salinarum	archaeon	`GCF_000015585.1_ASM1558v1`	extreme halophile (GC3/acid control)
SARS-CoV-2	virus	`GCF_009858895.2_ASM985889v3`	small-sample guard control (opt-in)

Table 1: Core 13 cellular RefSeq coding-sequence assemblies and the opt-in viral boundary control audited in the `v14.5` primary run.

3.2 True-frame parsing, one-gene-one-vote filtering, and genetic code audit

Each FASTA record is parsed in its annotated reading frame. To eliminate isoform inflation and alternative-splicing leakage, we enforce a strict **one-gene-one-vote** rule:

1. extract the canonical gene identifier from the FASTA header, prioritizing `[gene=...]`, then `[locus_tag=...]`, then `[protein_id=...]`;

2. strip transcript/isoform suffixes (`-RA`, `.1`, etc.) via `family_root()` when family-aware splitting is enabled;
3. retain only the single longest valid coding sequence per unique gene identifier;
4. trim the first (start) and last (stop) codons from every retained CDS record (`trim_start_stop=True`) so that initiation and termination context cannot masquerade as internal gene syntax, and discard records shorter than three internal codons.

At startup (`STAGE 00`), the pipeline executes a built-in unit test (`self_test_invariants()`) that verifies exact DNA codon-to-amino-acid mappings for eight NCBI translation tables: Table 1 (Standard), Table 2 (Vertebrate Mitochondrial: `AGA/AGG` → `*`, `ATA` → `M`, `TGA` → `W`), Table 3 (Yeast Mitochondrial: `CTN` → `T`, `ATA` → `M`, `TGA` → `W`), Table 4 (Mold/Protozoan Mitochondrial and *Mycoplasma*: `TGA` → `W`), Table 5 (Invertebrate Mitochondrial: `ATA` → `M`, `TGA` → `W`, `AGA/AGG` → `S`), Table 6 (Ciliate Nuclear: `TAA/TAG` → `Q`, `TGA` remains `*`), Table 11 (Bacterial/Archaeal/Plastid, sharing the standard sense/stop map of Table 1), and Table 12 (Alternative Yeast Nuclear: `CTG` → `S`), alongside verifying the 32 reverse-complement mirror pairs and (in `v14.6`) a synthetic 3-exon / 2-intron `join(...)` splicing reconstruction invariant. During CDS loading, `cds_units()` tracks `[transl_table=N]` per record, sets the dominant genetic code for the assembly, and filters out any discordant records (such as embedded mitochondrial sequences inside nuclear FASTA files) so that mixed genetic codes never contaminate a taxon's transition counts. Across all eight audits (the 54 eukaryotic nuclear and prokaryotic cellular assemblies plus SARS-CoV-2), every dominant assembly header resolves to Table 1 (`exact_map`), resulting in `code_mismatch skips = 0` across all transfer matrices (Table 12 is supported in code for *Candida albicans* when explicitly tagged in FASTA headers; see Section 7.2).

3.3 Family-aware train/test split and directional small-sample guard

For each species, up to `max_cds_records=8000` deduplicated genes with at least `min_trans=100` internal codon transitions are partitioned into an 80% training set and a 20% held-out test set using a family-aware group split (`--family-split`, `seed=20261014`). No codon transition is ever counted across gene boundaries, and training counts are never evaluated as test data.

Because estimating a 64×64 conditional transition matrix (4,096 cells) requires sufficient gene diversity and transition volume, `v14.5` enforces an automatic small-sample guard with thresholds `min_transfer_units=20` and `min_transfer_transitions=50000`. For within-species claims (`within_sample_ok`), both the training split and the held-out test split of a species must meet both thresholds; otherwise the assembly is flagged as `SMALL_SAMPLE`. For a directed cross-species transfer pair ($S \to T$), where models are trained on source S's training split and evaluated on target T's held-out test split, the guard `is_small_sample(S, T)` checks that source S's *training* split and target T's *test* split each satisfy ≥ 20 units and $\geq 50{,}000$ transitions; pairs failing this directional check are automatically skipped (`skipped_small_sample`).

3.4 Empirical codon-bigram model

Let $a, b \in \{0, \ldots, 63\}$ index the 64 sense and stop codons in lexicographic order over $\{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{T}\}^3$. From the training genes only, we accumulate transition counts $N_{ab} = \#\{i : c_i = a,\ c_{i+1} = b\}$. The Laplace/Dirichlet-smoothed empirical transition probability is

$$\hat{p}_{\mathrm{emp}}(b \mid a) = \frac{N_{ab} + \alpha}{\sum_{c=0}^{63} N_{ac} + 64\alpha}, \qquad \alpha = 0.1.$$

For a held-out test set containing T total codon transitions, the per-transition log-likelihood under any conditional model $q(b \mid a)$ is

$$\mathrm{LL}(q) = \frac{1}{T} \sum_{\text{test genes}} \sum_{i} \log q(c_{i+1} \mid c_i),$$

measured in nats per codon transition (natural logarithms).

4 Hierarchy of null models and geometric form invariants

4.1 H0–H3b: Analytical compositional, amino-acid, and sequential nulls

- **H0 (Codon marginal):** $q_0(b \mid a) = \hat{p}_{\text{marg}}(b)$, absorbing global codon frequencies without sequential order.
- **H1 (Boundary nucleotide):** $q_1(b \mid a) = \hat{p}(b \mid n_3(a))$, conditioning on the third nucleotide of the preceding codon.
- **H2 (Amino-acid syntax + marginal synonymous usage):** $q_2(b \mid a) = \hat{p}(\text{aa}(b) \mid \text{aa}(a))\, \hat{p}(b \mid \text{aa}(b))$, absorbing protein-level dipeptide succession and global synonymous bias.
- **H2b (Amino-acid syntax + codon-pair usage [16, 17]):**

$$q_{2b}(b \mid a) = \hat{p}(\text{aa}(b) \mid \text{aa}(a)) \, \frac{N_{ab} + \alpha}{\sum_{c:\, \text{aa}(c)=\text{aa}(b)} (N_{ac} + \alpha)},$$

  absorbing both amino-acid bigram syntax and synonymous codon preference conditioned on the preceding codon.
- **H3 (Folded dinucleotide composition):** For $a = a_1a_2a_3$ and $b = b_1b_2b_3$, $q_3(b \mid a) = \hat{p}(b_1 \mid a_3)\hat{p}(b_2 \mid b_1)\hat{p}(b_3 \mid b_2)$, absorbing 1st-order nucleotide Markov effects (such as CpG dinucleotide suppression) folded into codon space.
- **H3b (Folded nucleotide trigram, order-2):**

$$q_{3b}(b \mid a) = \hat{p}(b_1 \mid a_2a_3)\, \hat{p}(b_2 \mid a_3b_1)\, \hat{p}(b_3 \mid b_1b_2),$$

  absorbing all nucleotide Markov dynamics of order ≤ 2 across codon boundaries.

For each analytical null $k \in \{0, 1, 2, 2b, 3, 3b\}$, we report the held-out excess log-likelihood $d_k = \text{LL}(\hat{p}_{\text{emp}}) - \text{LL}(q_k)$.

4.2 H4: Block-shuffle local order control

For each held-out gene $c_1, \ldots, c_n$, the sequence is partitioned into contiguous blocks of length L codons ($L \in \{3, 10, 30, 100\}$), the blocks are randomly permuted, and any remainder $< L$ is kept at the tail. This preserves all internal correlations at scales $< L$ while destroying inter-block order at scales $\geq L$. Scored under the training empirical model $\hat{p}_{\text{emp}}$,

$$\Delta_{H4}(L) = \text{LL}_{\text{real}} - \text{LL}^{(L)}_{\text{block_shuffle}}.$$

The strict within-species and transfer gate is evaluated at $L = 30$ codons (90 nt), with $L = 100$ (300 nt) reported as a long-range diagnostic.

4.3 H5: Amino-acid-preserving H2b surrogate

For each held-out test gene, we fix the first codon c_1 and the exact amino-acid sequence $\text{aa}(c_2), \ldots, \text{aa}(c_n)$. At each step $i \geq 2$, a synonymous codon s_i encoding $\text{aa}(c_i)$ is drawn from the training H2b conditional distribution given the previously sampled codon s_{i-1}. Both the real gene and the surrogate are scored under $\hat{p}_{\text{emp}}$:

$$\Delta_5 = \text{LL}_{\text{real}} - \text{LL}_{\text{H2b_surrogate}}.$$

A positive Δ_5 demonstrates that real genes are more probable under the empirical transition model than synthetic sequences that already respect the exact protein sequence and training codon-pair conditional probabilities.

4.4 H6: Gene-specific synonymous multiset shuffle

To control for gene-by-gene heterogeneity in codon usage bias (e.g. highly expressed versus lowly expressed genes), H6 permutes synonymous codons exclusively within the positions of each amino acid *inside each individual held-out gene*. This preserves both the exact protein sequence and the exact per-gene synonymous codon counts while destroying sequential order:

$$\Delta_6 = \mathrm{LL}_{\mathrm{real}} - \mathrm{LL}_{\mathrm{syn_shuffle}}.$$

4.5 H7: Mirror-pair-preserving shuffle (Chargaff codon-parity control) and geometric form metrics

Let $\pi(c)$ denote the reverse-complement codon of c. Since no triplet is self-complementary, the 64 codons partition into 32 disjoint mirror pairs $M_p = \{c, \pi(c)\}$ $(p = 1, \ldots, 32)$, which form the basis of the single-stranded codon parity rule of Perez [6] (Sec. 2.2 therein). For each held-out gene, H7 identifies all positions belonging to each mirror pair M_p and randomly permutes the codons within those positions. Every gene's exact mirror-pair population vector $(|M_1|, \ldots, |M_{32}|)$ is preserved identically, while sequential order is destroyed:

$$\Delta_7 = \mathrm{LL}_{\mathrm{real}} - \mathrm{LL}_{\mathrm{mirror_shuffle}}.$$

A positive Δ_7 with a positive 95% bootstrap lower bound demonstrates that conditional codon syntax is orthogonal to static reverse-complement mirror symmetry.

To quantify the geometric transition from intronic *form* to exonic *program* in Section 6.5, `v14.6` also computes four explicit sequence-level invariants on each genomic compartment:

1. **Mirror-pair L_1 asymmetry:** $\mathrm{Mirror}_{L_1} = \frac{1}{2}\sum_{c=0}^{63} |f(c) - f(\pi(c))| \in [0, 1]$, measuring the departure of sense-strand codon frequencies $f(c)$ from exact Chargaff/Perez mirror symmetry $f(c) = f(\pi(c))$ (for reference, across all 947,803,867 frame-1 triplets of the whole single-stranded human genome in Table 3 of [6], where non-coding and intronic regions represent $\sim 98\%$ of DNA, odd- and even-sorted mirror codons sum to 474,337,193 and 473,466,674 from `TTT`/`AAA` down to `TCG`/`CGA`, corresponding to a near-zero whole-genome baseline $\mathrm{Mirror}_{L_1} \approx 0.0009$);
2. **Unordered 32-mirror-pair entropy H_{mirror} (≤ 5 bits) and 64-codon Shannon entropy H_{codon} (≤ 6 bits);**
3. **Markov flow time-reversal asymmetry:** $\mathrm{FlowAsym}_F = \|B - B^{\mathsf{T}}\|_F / \|B\|_F$, measuring directional irreversibility of the 64×64 codon bigram count matrix B;
4. **Internal stop-codon density (**Stops/100**):** in-frame stop triplets per 100 codons.

5 Statistical criteria and two-scale transfer metrics

5.1 Within-species strict stack with full bootstrap CIs (`v14.5`)

A species passes the `STRICT` within-species stack if and only if it satisfies the sample-size guard (≥ 20 train/test units and $\geq 50{,}000$ train/test transitions) and meets all seven point thresholds:

$$d_{2b} > 0.002, \qquad d_3 > 0.002, \qquad d_{3b} > 0.002, \tag{1}$$

$$\Delta_{H4}(30) > 0.0005, \qquad \Delta_5 > 0.001, \qquad \Delta_6 > 0.001, \qquad \Delta_7 > 0.001, \tag{2}$$

and the lower bound of the 95% gene-level bootstrap confidence interval (2.5th percentile) is strictly positive (> 0) for all seven quantities: d_{2b}, d_3, d_{3b}, $\Delta_{H4}(30)$, Δ_5, Δ_6, and Δ_7. In `v14.5 full` mode, point estimates use 8 block-shuffle replicates per gene for H4, 3 surrogate draws per gene for H5, 4 synonymous shuffles per gene for H6, and 4 mirror-pair shuffles per gene for H7. Bootstrap CIs resample held-out test genes with replacement using 120 replicates for d_0–d_{3b}, 40 replicates (with

2 inner block shuffles per gene per replicate fixed in `bootstrap_h4()`) for $\Delta_{H4}(30)$, 20 replicates (with `inner_reps=2` surrogate draws per gene per replicate via `--h5-bootstrap-inner-reps`) for Δ_5, 20 replicates for Δ_6, and 20 replicates for Δ_7.

5.2 Exhaustive coarse cross-species transfer and `coarse_clean_0p02`

The coarse transfer audit is exhaustive over all $N(N-1)$ ordered pairs of loaded species (S, T) in a given run. For each ordered pair sharing the same translation table and passing the directional small-sample guard, we score target T's held-out test genes under source S's training models. The pair passes the basic coarse gate (`TRANSFER OK +H4`) if $d_{2b}^{\text{transfer}} > 0.001$, $d_{3}^{\text{transfer}} > 0.001$, $d_{3b}^{\text{transfer}} > 0.001$, and $\Delta_{H4}^{\text{transfer}}(30) > 0.0005$. It passes the macro-clade synapomorphy tier (`coarse_clean_0p02`, tagged `[CLEAN>0.02]`) if additionally

$$d_{2b}^{\text{transfer}}(S \to T) > 0.020.$$

5.3 Focal fine H5-transfer conservation and sensitivity tiers

Whereas the coarse transfer audit is exhaustive over all $N(N-1)$ directed pairs, the fine H5-transfer audit is *focal*: because Monte Carlo surrogate generation across every possible pair of 54 species is computationally intensive, the pipeline evaluates pre-specified lists of phylogenetically informative conservation and control contrasts (`DEFAULT_H5_TRANSFER_PAIRS` [40 core pairs], `OPTIN_H5_TRANSFER_PAIRS` [8 viral pairs], and `EXTENDED_H5_TRANSFER_PAIRS` [104 extended pairs]; complete audit-by-audit inventory in Appendix E). In the core+virus audit (`v14_5_paper_final`), this focal list comprises 40 core pairs plus 8 opt-in SARS-CoV-2 pairs; in the 12-mammal radiation audit (`v14_5_ext_mammals`), it comprises 34 directed mammalian pairs whose both endpoints are loaded (6 core + 28 extended).

For each focal pair $(S \to T)$ passing the directional small-sample guard, `v14.5` generates amino-acid-preserving H2b surrogates for target T's test genes using *source S's* training H2b sampler (`h5_transfer_reps=12` in `paper_final`, `4` in extended runs), and scores both real target genes and source-generated surrogates under *target T's* empirical model $\hat{p}_{\text{emp}}^{(T)}$. Let $\Delta_5{}_T^{\text{transfer}}$ be the resulting excess log-likelihood, and let $\Delta_5{}_T^{\text{within}}$ be target T's native within-species H5 delta. We define the conservation gap and ratio:

$$\text{gap}_{S\to T} = \Delta_5{}_T^{\text{transfer}} - \Delta_5{}_T^{\text{within}}, \qquad \text{ratio}_{S\to T} = \frac{\Delta_5{}_T^{\text{transfer}}}{\Delta_5{}_T^{\text{within}}}.$$

To ensure that compositional hardware mismatches cannot be misclassified as shared program syntax, `v14.5` assigns each evaluated pair to one of four mutually exclusive status categories:

1. `H5-CONSERVED`**:** satisfies $\Delta_5{}_T^{\text{within}} > 0.001$, $\Delta_5{}_T^{\text{transfer}} > 0$, $|\text{gap}_{S\to T}| \le 0.005$, and $0.80 \le \text{ratio}_{S\to T} \le 1.25$ (with `artifact_candidate=False`).
2. `GC/aa artifact candidate`**:** not `H5-CONSERVED`, and exhibits a large compositional distance between source and target, $|\text{GC3}_S - \text{GC3}_T| > 0.12$ or $|\text{acid}_S - \text{acid}_T| > 0.08$ (`artifact_candidate=True`).
3. `H5-NEAR-MISS` **(sensitivity tier):** neither `H5-CONSERVED` nor a `GC/aa artifact candidate` (`not conserved and not artifact`), and satisfies $\Delta_5{}_T^{\text{within}} > 0.001$, $\Delta_5{}_T^{\text{transfer}} > 0$, $0.80 \le \text{ratio}_{S\to T} \le 1.25$, and $0.005 < |\text{gap}_{S\to T}| \le 0.012$.
4. `not conserved` **(true divergence):** fails conservation and near-miss criteria despite matched composition ($\Delta\text{GC3} \le 0.12$, $\Delta\text{acid} \le 0.08$).

Across all eight reported audits, no pair with large compositional distance ($|\Delta\text{GC3}| > 0.12$ or $|\Delta\text{acid}| > 0.08$) ever satisfied the raw numerical `H5-CONSERVED` inequalities, so evaluating the compositional guard before or after `H5-CONSERVED` is empirically identical. Every non-artifact

H5-CONSERVED or H5-NEAR-MISS pair across all runs had $|\Delta\text{GC3}| \leq 0.106$ and $|\Delta\text{acid}| \leq 0.007$; within the 12-mammal focal matrix the corresponding maxima are $|\Delta\text{GC3}| = 0.040$ and $|\Delta\text{acid}| = 0.004$. Furthermore, one extreme-composition contrast in the dataset—*Leishmania major* → *Trypanosoma brucei* in ext_protists ($\Delta\text{GC3} = 0.232$, $\Delta\text{acid} = 0.012$, gap $= -0.005636$, ratio $= 0.819$)—fell inside the raw numerical near-miss window and was intercepted by the compositional guard (artifact_candidate=True) prior to H5-NEAR-MISS assignment, directly illustrating the practical necessity of the compositional hardware guard.

6 Results

6.1 Primary audit (v14_5_paper_final): 13 cellular taxa and viral boundary control

Table 2 and Table 3 report the primary v14.5 full audit (seed=20261014, --family-split). All thirteen cellular species pass every criterion of the STRICT stack, with all 95% bootstrap confidence intervals bounded strictly above zero, including the newly bootstrapped Δ_6 and Δ_7 controls (exact CIs in Appendix Table 14).

Species	Clade	GC3	Acid	d_{2b}	d_3	d_{3b}	$\Delta_{H4}(30)$	Δ_5	Δ_6	Δ_7
Drosophila	insect	0.6330	0.1159	+0.033840	+0.184504	+0.156669	+0.003888	+0.066710	+0.093968	+0.068364
Mouse	mammal	0.5738	0.1157	+0.053629	+0.128007	+0.107767	+0.004844	+0.113807	+0.129894	+0.112315
Arabidopsis	plant	0.4176	0.1211	+0.024823	+0.125765	+0.102762	+0.002551	+0.051293	+0.070959	+0.054444
Zebrafish	vertebrate	0.5358	0.1225	+0.048775	+0.134910	+0.108936	+0.004881	+0.101286	+0.130481	+0.110552
C. elegans	nematode	0.3948	0.1169	+0.027086	+0.125589	+0.095408	+0.003115	+0.056219	+0.079755	+0.065205
E. coli	bacterium	0.5620	0.1094	+0.044580	+0.270848	+0.215570	+0.005523	+0.095551	+0.154321	+0.100070
M. jannaschii	archaeon	0.4205	0.1133	+0.063336	+0.266198	+0.223715	+0.006391	+0.133646	+0.184275	+0.135903
B. subtilis	bacterium	0.4465	0.1248	+0.041548	+0.196092	+0.154256	+0.004163	+0.084581	+0.126240	+0.097930
H. salinarum	archaeon	0.8630	0.1360	+0.035991	+0.358761	+0.298201	+0.004742	+0.083182	+0.103297	+0.080797
Yeast	fungus	0.3776	0.1242	+0.010547	+0.138061	+0.122536	+0.001227	+0.024270	+0.028782	+0.024900
Chimp	mammal	0.5851	0.1159	+0.048665	+0.141619	+0.119933	+0.004602	+0.104349	+0.126270	+0.109523
Human_T2T	mammal	0.5791	0.1175	+0.047474	+0.140074	+0.119127	+0.004608	+0.098854	+0.125331	+0.109692
Chicken	vertebrate	0.5741	0.1185	+0.044844	+0.141323	+0.118544	+0.004811	+0.090364	+0.127790	+0.112165
SARS-CoV-2	virus	0.2822	0.0979	−0.067011	−0.417539	−0.416335	+0.000724	−0.448115	+0.093021	+0.050864

Table 2: Within-species held-out log-likelihood excesses (in nats per codon transition) from the v14_5_paper_final audit. All 13 cellular species pass the STRICT stack; SARS-CoV-2 fails the small-sample guard (SMALL_SAMPLE) and multiple analytical/H4/H5/H6 bootstrap criteria.

Species	Train units	Test units	Train transitions	Test transitions	Verdict
Drosophila	6009	1496	3,261,810	852,481	STRICT
Mouse	6147	1536	3,453,102	851,610	STRICT
Arabidopsis	5904	1473	2,520,418	623,252	STRICT
Zebrafish	6251	1572	3,618,904	898,735	STRICT
C. elegans	5862	1459	2,591,840	646,698	STRICT
E. coli	3011	752	1,021,490	252,839	STRICT
M. jannaschii	1547	386	524,118	129,962	STRICT
B. subtilis	2871	717	981,205	247,148	STRICT
H. salinarum	1800	450	660,112	168,116	STRICT
Yeast	4539	1134	2,310,442	583,505	STRICT
Chimp	6210	1552	3,512,088	867,394	STRICT
Human_T2T	6159	1539	3,589,410	898,022	STRICT
Chicken	6250	1562	3,641,200	905,030	STRICT
SARS-CoV-2	9	2	2,512	7,213	SMALL_SAMPLE

Table 3: Held-out sample sizes in v14_5_paper_final after one-gene-one-vote deduplication, start/stop trimming, and family-aware 80/20 splitting.

The viral control (SARS-CoV-2) illustrates why the small-sample guard and bootstrap CIs are essential: its genome contains only 11 deduplicated CDS units, with over 70% of all codons concentrated in a single giant polyprotein (`ORF1ab`, $\sim$ 7,096 codons). When `ORF1ab` falls into the 2-unit test split (7,213 transitions), the nine smaller non-`ORF1ab` ORFs in the training split cannot populate a 64×64 transition matrix, causing the analytical deltas and Δ_5 to collapse to strongly negative values ($d_0 = -0.542114$ and $d_1 = -0.522790$ in the run log; $d_3 = -0.417539$, $d_{3b} = -0.416335$, $d_{2b} = -0.067011$, and $\Delta_5 = -0.448115$ in Table 2), while the H4 and H6 bootstrap CIs ($[-0.0236, +0.0029]$ and $[-0.0472, +0.1062]$) expose the positive point estimates as statistical noise. The guard automatically excludes SARS-CoV-2 from cellular claims and cross-species transfer matrices (26 coarse skips, 8 H5 skips). The difference between 26 coarse skips and 8 H5 skips reflects the two-scale audit design (§5.3): coarse transfer is evaluated exhaustively over all $14 \times 13 = 182$ directed pairs (26 involving SARS-CoV-2, leaving 156 cellular pairs), whereas fine H5-transfer evaluates 40 core cellular pairs plus the 8 pre-specified viral contrasts in `OPTIN_H5_TRANSFER_PAIRS["SARS-CoV-2"]`.

6.2 Characteristic scale of conditional syntax: $L = 30$ versus $L = 100$

Across all 13 core cellular species, block-shuffle excess log-likelihood decays monotonically with block length L. At $L = 3$ codons, $\Delta_{H4}(3)$ ranges from +0.0134 (Yeast) to +0.0696 (*M. jannaschii*); at $L = 10$, from +0.0041 to +0.0207; at the strict gate $L = 30$ (90 nt), every species remains robustly positive with tight 95% CIs (+0.00123 in Yeast to +0.00639 in *M. jannaschii*). By $L = 100$ codons (300 nt), however, the point estimates are strongly attenuated (decaying by factors of approximately 2.3 to 8.7 among positive residuals) or turn negative: Drosophila (+0.000657), Mouse (+0.000975), Arabidopsis (+0.000465), Zebrafish (+0.001144), *C. elegans* (−0.000455), *E. coli* (−0.006423), *M. jannaschii* (−0.003268), *B. subtilis* (+0.000477), *H. salinarum* (−0.004853), Yeast (−0.000766), Chimp (−0.000253), Human_T2T (+0.000939), and Chicken (+0.002094). Biophysically, the $L = 30$ codon scale (90 nt, encoding 30 amino acids) matches the length of the ribosomal exit tunnel ($\sim$ 30–40 amino acids) and early co-translational nascent-polypeptide folding windows, rather than the minimal ribosomal protected footprint ($\sim$ 9–10 codons) or whole-gene compositional drift, whereas the $L = 100$ codon ($\sim$ 300 nt) boundary where first-order Markovian transition syntax attenuates marks the transition to the mesoscopic $\sim$ 100-codon (312 nt) integration window over which cumulative genomics/proteomics Master Code coupling curves operate [4, 7, 10]. Remarkably, in discrete transition-graph terms, a contiguous block of $L = 30$ codons (90 nt) spans **exactly** $L - 1 = 29$ **first-order codon-to-codon Markov transitions** ($c_1 \to c_2, \ldots, c_{29} \to c_{30}$) and **exactly** $3L - 1 = 89$ **nucleotide-to-nucleotide steps** ($n_1 \to n_2, \ldots, n_{89} \to n_{90}$): **29** is the 8th Lucas number ($2, 1, 3, 4, 7, 11, 18, \mathbf{29}, 47, 76$) and **89** is the 11th Fibonacci number ($1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \mathbf{89}$)—the exact characteristic resonance scales at which Perez [11] demonstrated forward/backward supracode symmetry in natural genomes and its breakdown in synthetic constructs (such as synthetic yeast *synXII* `PCRTags` at Lucas 29 and *Mycoplasma / Caulobacter* at Fibonacci 89).

6.3 Two-scale transfer in the core panel: vertebrate synapomorphy and fine asymmetry

Table 4 displays the coarse transferred conditional excess d_{2b}^{transfer} among the five core vertebrates in `v14_5_paper_final`. All 20/20 directed pairs pass `TRANSFER OK +H4` and exceed the macro-layer threshold $d_{2b} > 0.020$ (range +0.03175 to +0.05216), whereas not a single non-vertebrate pair in the 156-pair core matrix reaches 0.020 (the highest non-vertebrate transfer is Arabidopsis → Zebrafish at $d_{2b} = +0.01691$).

At the fine scale (Table 5), `v14_5_paper_final` (with 12 surrogate repetitions per transfer pair) resolves three distinct regimes:

Source ↓ / Target →	Human_T2T	Chimp	Mouse	Zebrafish	Chicken
Human_T2T	—	+0.04862	+0.05216	+0.03443	+0.04122
Chimp	+0.04754	—	+0.05198	+0.03455	+0.04143
Mouse	+0.04577	+0.04667	—	+0.03570	+0.03989
Zebrafish	+0.03243	+0.03327	+0.04005	—	+0.03175
Chicken	+0.04424	+0.04547	+0.04871	+0.03601	—

Table 4: Coarse cross-species transfer (d_{2b}^{transfer}) among the five core vertebrates in `v14_5_paper_final`. All 20 directed pairs pass `+H4` and exceed $d_{2b} = 0.020$ (`[CLEAN>0.02]`).

1. **Hominid bidirectional conservation:** Human_T2T → Chimp (gap = +0.000151, ratio = 1.001) and Chimp → Human_T2T (gap = +0.001381, ratio = 1.014) pass `H5-CONSERVED` with virtually zero divergence.
2. **Near-miss sensitivity tier and directional rodent asymmetry:** In the primary core audit, Human_T2T → Mouse (gap = −0.006483, ratio = 0.943), Chimp → Mouse (gap = −0.005884, ratio = 0.948), and Human_T2T → Chicken (gap = +0.006091, ratio = 1.067) lie in the `H5-NEAR-MISS` tier ($|\text{gap}| \leq 0.012$). In contrast, the reverse directions Mouse → Human_T2T (gap = +0.020633, ratio = 1.209) and Mouse → Chimp (gap = +0.019942, ratio = 1.191) exhibit positive gaps that are 3.2–3.4-fold larger in magnitude and fail both tiers despite a very small GC3 distance of 0.005–0.011.
3. **True lineage divergence versus compositional hardware mismatch:** Zebrafish (ΔGC3 = 0.043), Drosophila (ΔGC3 = 0.054), and *E. coli* (ΔGC3 = 0.017) fail fine conservation against Human_T2T without triggering the composition guard, demonstrating genuine divergence of their synonymous grammars. Meanwhile, all 22 contrasts with $|\Delta\text{GC3}| > 0.12$ (led by *H. salinarum*, ΔGC3 = 0.284 to Human and 0.443 to *M. jannaschii*) are automatically isolated as `GC/aa artifact candidates`.

Source → Target	$\Delta_5{}_T^{\text{transfer}}$	$\Delta_5{}_T^{\text{within}}$	gap	ratio	GC3 dist	Status (`v14.5`)
Human_T2T → Chimp	+0.104499	+0.104349	+0.000151	1.001	0.006	`H5-CONSERVED`
Chimp → Human_T2T	+0.100234	+0.098854	+0.001381	1.014	0.006	`H5-CONSERVED`
Human_T2T → Mouse	+0.107325	+0.113807	−0.006483	0.943	0.005	`H5-NEAR-MISS`
Chimp → Mouse	+0.107923	+0.113807	−0.005884	0.948	0.011	`H5-NEAR-MISS`
Mouse → Human_T2T	+0.119486	+0.098854	+0.020633	1.209	0.005	`not conserved`
Mouse → Chimp	+0.124290	+0.104349	+0.019942	1.191	0.011	`not conserved`
Human_T2T → Chicken	+0.096455	+0.090364	+0.006091	1.067	0.005	`H5-NEAR-MISS`
Chicken → Human_T2T	+0.111508	+0.098854	+0.012654	1.128	0.005	`not conserved`
Human_T2T → Zebrafish	+0.123892	+0.101286	+0.022606	1.223	0.043	`not conserved`
Zebrafish → Human_T2T	+0.146912	+0.098854	+0.048059	1.486	0.043	`not conserved`
Drosophila → Human_T2T	+0.183484	+0.098854	+0.084630	1.856	0.054	`not conserved`
E. coli → Human_T2T	+0.264421	+0.098854	+0.165568	2.675	0.017	`not conserved`
H. salinarum → Human_T2T	+0.066176	+0.098854	−0.032677	0.669	0.284	`GC/aa artifact candidate`
Human_T2T → H. salinarum	+0.786039	+0.083182	+0.702857	9.450	0.284	`GC/aa artifact candidate`

Table 5: Selected fine H5-transfer conservation results from `v14_5_paper_final` (`h5_transfer_reps=12`).

6.4 Extended biosphere validation across 54 cellular taxa and Primate–Rodent directional asymmetry

To test whether the core findings depend on taxon sampling, we executed seven additional domain-specific full audits across the 41 extended RefSeq assemblies (`v14_5_ext_mammals`, `v14_5_ext_vert`, `v14_5_ext_inv`, `v14_5_ext_plants`, `v14_5_ext_fungi`, `v14_5_ext_protists`, and `v14_5_ext_bact`, making eight full multi-species audits in total including `paper_final`). Table 6 summarizes the global results across all seven extended domain audits, Appendix Table 15 reports the complete within-species metrics for every extended assembly, Appendix Table 16 reports the focal 34-pair

evaluated H5-transfer matrix used for the fine-scale mammalian analysis, and Appendix Table 17 provides the complete focal H5-transfer accounting across all eight audits.

Audit panel	Taxa included	GC3 range	STRICT	Coarse evaluable pairs	CLEAN>0.02	Skips	Fine H5-CONSERVED highlights
ext_mammals	12 mammals (6 orders)	0.5714–0.6180	12 / 12	132 / 132	132 (100%)	0	18 pairs (intra-ordinal cores + Cat/Horse→Human bridges)
ext_vert	5 vertebrates (4 classes)	0.4753–0.6411	5 / 5	20 / 20	20 (100%)	0	*Takifugu* → Zebrafish (gap = +0.0029)
ext_inv	4 insects, 2 nematodes	0.3303–0.6977	6 / 6	30 / 30	2 (6.7%)	0	*Anopheles* → *Drosophila* (gap = +0.0032)
ext_plants	6 land plants, 1 green alga	0.3637–0.8143	7 / 7	42 / 42	8 (19.0%)	0	*Populus* → *Arabidopsis* (gap = +0.0048)
ext_fungi	5 yeast/filamentous fungi	0.2759–0.6463	5 / 5	20 / 20	0 (0.0%)	0	None (diverged / GC-separated)
ext_protists	5 parasitic/free protists	0.1416–0.7466	4 / 5*	16 / 20	0 (0.0%)	4	None (extreme GC3 divergence)
ext_bact	13 bacteria, 2 archaea	0.1733–0.9177	14 / 15*	196 / 210	4 (2.0%)	14	None (isolated / GC-separated)

Table 6: Summary of the seven extended domain-specific biosphere audits in v14.5 (code_mismatch skips = 0 in all panels; GC3 range reports held-out test-split GC3 within each run, with exact per-taxon values recorded in the run JSON reports). Fine H5-CONSERVED highlights refer to the focal H5-transfer audit, not to all coarse evaluable directed pairs. (*In ext_protists and ext_bact, *Leishmania major* and *Staphylococcus aureus* had systematic locus_tag prefixes grouped into a large training cluster by family_root(), leaving 83 test units / 44,196 transitions and 38 test units / 13,280 transitions respectively. Both achieved strictly positive point deltas and 95% bootstrap CIs on all 7 nulls, but fell below the 50,000 test-transition guard. Because the directional transfer guard requires $\geq$ 50,000 training transitions in the source and $\geq$ 50,000 test transitions in the destination, this produced 4 destination-side small-sample skips in ext_protists [leaving $20 - 4 = 16$ coarse evaluable pairs] and 14 destination-side skips in ext_bact [leaving $210 - 14 = 196$ coarse evaluable directed pairs, of which 4 pass CLEAN>0.02, i.e. 2.0% of all 196 evaluable pairs or 2.2% when restricted to the $14 \times 13 = 182$ pairs whose both endpoints clear the guard].)

Three major biological findings emerge from the 54-taxon expansion:

1. Ubiquitous within-species conditional syntax across the sampled biological GC3 spectrum. All 52 cellular species meeting the within-species sample-size guard pass the full STRICT stack on $d_{2b}, d_3, d_{3b}, \Delta_{H4}(30), \Delta_5, \Delta_6$, and Δ_7. The two small-split assemblies, *Leishmania major* and *Staphylococcus aureus*, do not receive a formal STRICT verdict because their held-out test splits fall below the 50,000-transition guard, but both exhibit strictly positive point estimates and positive 95% bootstrap lower bounds across all seven quantities (Appendix Table 15). At the extreme AT-rich pole, the social amoeba *Dictyostelium discoideum* (GC3 = 0.1416), the malaria parasite *Plasmodium falciparum* (GC3 = 0.1726), and the gut pathogen *Clostridioides difficile* (GC3 = 0.1733) all maintain robust medium-range conditional syntax ($\Delta_{H4}(30) = +0.00576, +0.00283, +0.00179$). At the extreme GC-rich pole, the thermophile *Thermus thermophilus* (GC3 = 0.9177) and *Pseudomonas aeruginosa* (GC3 = 0.8803) exhibit the strongest conditional syntax in the dataset ($d_{2b} = +0.12137$, $\Delta_{H4}(30) = +0.01438$, $\Delta_5 = +0.31045$, $\Delta_6 = +0.38366$, $\Delta_7 = +0.28918$ in *T. thermophilus*).

2. The 12-mammal radiation and reproducible Euarchontoglires directional asymmetry. In v14_5_ext_mammals, we audited 12 mammalian genomes across six orders: Primates (*Homo sapiens*, *Pan troglodytes*, *Gorilla gorilla*, *Macaca mulatta*), Rodentia (*Mus musculus*, *Rattus norvegicus*), Lagomorpha (*Oryctolagus cuniculus*), Carnivora (*Canis lupus familiaris*, *Felis catus*), Artiodactyla (*Bos taurus*, *Sus scrofa*), and Perissodactyla (*Equus caballus*). Every one of the **132/132** directed pairs passes coarse_clean_0p02 ($d_{2b} \in [+0.03621, +0.05505]$).

Crucially, adding a second rodent (*Rattus norvegicus*) and two additional primates (*Gorilla gorilla* and the Old World monkey *Macaca mulatta*) supports the interpretation that the Mouse asymmetry observed in the core run is not a single-species anomaly, but a reproducible **Euarchontoglires-level directional pattern** (Table 7 and Appendix Table 16):

- **Human-centered Primate symmetry:** Human $\leftrightarrow$ Chimp, Human $\leftrightarrow$ Gorilla, and Human $\leftrightarrow$ Macaque are all bidirectionally H5-CONSERVED ($|\text{gap}| \leq 0.00357$, ratio $\in$

$[0.984, 1.037]$).

- **Intra-Rodent symmetry:** Mouse $\leftrightarrow$ Rat is bidirectionally `H5-CONSERVED` (gap = +0.00128 and +0.00254, ratio = 1.011 and 1.022). Both rodents share an identically elevated internal H5 baseline (+0.1134 in Mouse, +0.1143 in Rat vs. +0.0969– + 0.1030 in Primates).
- **Intra-Laurasiatheria conservation and Laurasiatheria-to-Human bridges:** Within Laurasiatheria, Cow $\leftrightarrow$ Pig is bidirectionally `H5-CONSERVED` (gap = +0.00287 and +0.00066), and Dog $\rightarrow$ Cat is `H5-CONSERVED` (gap = −0.00415, reverse in `H5-NEAR-MISS` at +0.00734). Across superorders, Cat $\rightarrow$ Human (−0.00018) and Horse $\rightarrow$ Human (−0.00420) pass `H5-CONSERVED`, with reverse directions and Dog/Cow/Pig/Rabbit $\rightarrow$ Human in `H5-NEAR-MISS` (Appendix Table 16).
- **Directional Primate $\rightarrow$ Rodent conservation vs. Rodent $\rightarrow$ Primate overshoot in focal pairs:** All five examined focal Primate $\rightarrow$ Rodent pairs in `ext_mammals` pass `H5-CONSERVED` (gap $\in [-0.00494, -0.00316]$), whereas all five reverse Rodent $\rightarrow$ Primate pairs fail with a uniform positive overshoot (gap $\in [+0.01709, +0.02072]$, ratio $\in [1.166, 1.214]$) at virtually zero GC3 distance (ΔGC3 = 0.000–0.008 in `ext_mammals`; ≤ 0.011 including the core run).

The strict $|\text{gap}| \leq 0.005$ boundary is sensitive for the borderline Human/Chimp $\rightarrow$ Mouse pairs: due to differences in surrogate repetition count (`h5_transfer_reps=12` in `paper_final` vs. `4` in `ext_mammals`) and the RNG stream order across differently sized panels, these two pairs fall into `H5-NEAR-MISS` (gap = −0.00648 and −0.00588) in `paper_final` but cross into `H5-CONSERVED` (gap = −0.00494 and −0.00316) in `ext_mammals`. The robust, run-invariant signal is not the binary threshold crossing of these borderline pairs, but the systematic sign and magnitude separation: primate-to-rodent gaps are consistently small and negative (−0.0065 to −0.0032), whereas rodent-to-primate gaps are uniformly large and positive (+0.0171 to +0.0207).

Forward pair ($A \rightarrow B$)	$\text{gap}_{A\rightarrow B}$	$\text{ratio}_{A\rightarrow B}$	Forward status	Reverse pair ($B \rightarrow A$)	$\text{gap}_{B\rightarrow A}$	$\text{ratio}_{B\rightarrow A}$	Reverse status	GC3 dist
Symmetric intra-ordinal mammalian cores (bidirectional `H5-CONSERVED`):								
Human_T2T → Chimp	−0.001473	0.986	`H5-CONSERVED`	Chimp → Human_T2T	+0.002699	1.028	`H5-CONSERVED`	0.001
Human_T2T → Gorilla	−0.001607	0.984	`H5-CONSERVED`	Gorilla → Human_T2T	+0.003571	1.037	`H5-CONSERVED`	0.004
Human_T2T → Macaca	−0.000806	0.992	`H5-CONSERVED`	Macaca → Human_T2T	+0.001548	1.016	`H5-CONSERVED`	0.005
Mouse → Rattus	+0.001283	1.011	`H5-CONSERVED`	Rattus → Mouse	+0.002536	1.022	`H5-CONSERVED`	0.008
Bos_taurus → Sus_scrofa	+0.002873	1.033	`H5-CONSERVED`	Sus_scrofa → Bos_taurus	+0.000658	1.008	`H5-CONSERVED`	0.006
Directional Primate → Rodent conservation vs. Rodent → Primate overshoot in focal pairs:								
Human_T2T → Mouse	−0.004937	0.956	`H5-CONSERVED`	Mouse → Human_T2T	+0.019612	1.202	`not conserved`	0.007
Chimp → Mouse	−0.003155	0.972	`H5-CONSERVED`	Mouse → Chimp	+0.017089	1.166	`not conserved`	0.008
Human_T2T → Rattus	−0.004783	0.958	`H5-CONSERVED`	Rattus → Human_T2T	+0.020717	1.214	`not conserved`	0.001
Chimp → Rattus	−0.003423	0.970	`H5-CONSERVED`	Rattus → Chimp	+0.019007	1.184	`not conserved`	0.000
Macaca → Rattus	−0.004932	0.957	`H5-CONSERVED`	Rattus → Macaca	+0.018389	1.186	`not conserved`	0.003

Table 7: Fine H5-transfer symmetry within mammalian orders versus systematic Primate $\leftrightarrow$ Rodent directional asymmetry in the 12-mammal audit (`v14_5_ext_mammals`). Note that Chimp and *Rattus norvegicus* have an identical GC3 (ΔGC3 = 0.000) yet differ by a factor of 5.5$\times$ in directional H5 gap. All 34 focal evaluated pairs are listed in Appendix Table 16.

3. Five-taxon vertebrate bridge across four classes versus non-vertebrate taxonomic localization. In `v14_5_ext_vert`, adding the amphibian *Xenopus tropicalis* (GC3 = 0.4753) and the tetraodontiform pufferfish *Takifugu rubripes* (GC3 = 0.6411) to Human (Mammalia), Chicken (Aves), and Zebrafish (Cypriniformes, Actinopterygii) yields **20/20** directed pairs passing `coarse_clean_0p02` ($d_{2b} \in [+0.02185, +0.04402]$), showing that the coarse vertebrate layer spans mammals, birds, amphibians, and two teleost orders across a 16.6 percentage-point GC3 span. Furthermore, *Takifugu* $\rightarrow$ Zebrafish passes fine `H5-CONSERVED` (gap = +0.002870, ratio = 1.028), demonstrating fine-scale conservation between distantly related teleost orders, while transfers from Human, Chicken, and Zebrafish into *Xenopus* all land in `H5-NEAR-MISS` (gap $\in [+0.00781, +0.00981]$).

By contrast, in invertebrates (`ext_inv`), plants (`ext_plants`), fungi (`ext_fungi`), protists

(ext_protists), and prokaryotes (ext_bact), coarse_clean_0p02 and H5-CONSERVED occur mainly within narrower lower-rank clades: between congeneric nematodes (*C. elegans* ↔ *C. briggsae*, $d_{2b} = +0.02547/+0.02335$, with *C. elegans* → *C. briggsae* in H5-NEAR-MISS at gap = +0.00779), between dipteran insects (*Anopheles* → *Drosophila*, H5-CONSERVED, gap = +0.00321), within core eudicots (*Populus*, *Glycine*, *Solanum*, *Arabidopsis*, with 6 dicot CLEAN>0.02 pairs and *Populus* → *Arabidopsis* H5-CONSERVED at gap = +0.00478), within monocot grasses (*Zea mays* → *Oryza sativa*, $d_{2b} = +0.02111$), and within Enterobacteriaceae (*E. coli* ↔ *Salmonella enterica*, $d_{2b} = +0.04100/+0.03632$). The only exceptions outside these narrow clades are two high-GC prokaryotic CLEAN>0.02 transfers into *Pseudomonas aeruginosa* (*H. salinarum* → *P. aeruginosa*, $d_{2b} = +0.03172$, and *Mycobacterium tuberculosis* → *P. aeruginosa*, $d_{2b} = +0.02260$) and one coarse high-GC transfer from *Zea mays* into *Chlamydomonas reinhardtii* ($d_{2b} = +0.02191$, which reflects coarse GC3-driven transition overlap rather than fine H5 conservation).

6.5 Compartment-level control: Intron–Exon confinement, splice-junction phase, and the *TP53* splicing trajectory (v14.6)

If the conditional codon syntax established in the preceding results subsections reflects a genuine protein-coding *program* rather than a regional chromosomal artifact, a direct prediction is that it should obey a sharp compartment boundary inside split eukaryotic genes. Specifically, within the exact same genomic locus, the conditional syntax signal beyond order-2 nucleotide trigrams ($d_{3b} > 0$) and broken mirror symmetry ($\text{Mirror}_{L_1} \gg 0$) must be confined to **mature concatenated exons** and absent (or strongly negative) in **excised introns** and in the **whole unspliced pre-mRNA gene** across all reading frames.

To test this directly, super_codon_proof_v14_6.py (v14.6) autonomously fetches and parses official human NCBI RefSeqGene records (NG_*) across a 14-gene benchmark panel—comprising the complete *p*53 tumor-suppressor paralog family (*TP53*, *TP63*, *TP73*), the complete *RAS* proto-oncogene paralog family (*KRAS*, *NRAS*, *HRAS*), major multi-intron tumor suppressors (*BRCA1*, *PTEN*, *RB1*), *MYC*, high-expression housekeeping genes (*GAPDH*, *ACTB*), compact *HBB*, and high-GC *APOE*—and scores four structural states per locus against the human reference transition model built from the T2T-CHM13v2.0 CDS set (GCF_009914755.1, max_records=6000):

1. final_exons_only **(2/ Exons concatenated):** the mature spliced coding sequence in the true reading frame (start and stop codons trimmed);
2. full_gene_with_introns_f0 **(1/ Whole introns+exons, frame 0):** the entire unspliced pre-mRNA span from the first coding exon to the last coding exon read in frame 0;
3. full_gene_with_introns_3f **(3/ Whole introns+exons, all 3 frames):** the unspliced pre-mRNA span evaluated across reading frames 0, 1, and 2, recording the best frame d_{3b}^{best3f} so that intron-induced frame shifts cannot account for the loss of syntax;
4. introns_only **(4/ Excised introns only):** all excised coding introns concatenated.

Tables 8, 9, and 11 establish five compartment-level and sub-genic empirical regularities:

1. **14/14 compartment confinement of d_{3b}:** Across all 14 human loci—spanning intron-to-exon length ratios from 0.69× (*ACTB*) to 127.73× (*TP63*, 262,987 nt genomic span; 262,986 nt / 87,662 frame-0 triplets)—every spliced coding sequence has $d_{3b} > 0$ (range +0.0976 to +0.2686, mean +0.1721 nats/codon), whereas every excised intron concat has $d_{3b} < 0$ (range −0.4461 to −0.1435, mean contrast $\Delta d_{3b}(\text{Exons} - \text{Introns}) = +0.48182$ nats/codon) and every whole unspliced pre-mRNA locus has $d_{3b} < 0$ across all three reading frames (mean best-of-3-frames contrast +0.41617 nats/codon).
2. **Internal replication across both the *p*53 (*TP53*, *TP63*, *TP73*) and *RAS* (*KRAS*, *NRAS*, *HRAS*) paralog families:** All three human *p*53 family members (extending the 393 × 61 *TP53* mutability and Master Code hotspot prediction analysis of Perez [9] and Perez and Montagnier [10] to Markovian compartment dynamics) independently reproduce the transition from mirror-symmetric intronic form approaching the

Gene	RefSeqGene	Ex/In	I/E	d_{3b} (nats/codon)				Δd_{3b} contrast		Mirror$_{L_1}$		FlowAsym$_F$		Intron
symbol	accession	count	ratio	Exons	Whole f0	Best 3f	Introns	Ex−Best3f	Ex−In	Exons	Introns	Exons	Introns	Stops/100
TP53	NG_017013.2	10/9	4.91x	+**0.1117**	−0.3176	−0.2554	−0.2738	+0.3672	+0.3856	**0.3878**	0.0951	1.2353	1.0506	3.77
TP63	NG_007550.1	14/13	127.73x	+**0.0976**	−0.3871	−0.3798	−0.3865	+0.4774	+0.4841	**0.3549**	0.0311	1.1592	0.3568	6.08
TP73	NG_017035.2	13/12	25.54x	+**0.2291**	−0.1826	−0.1826	−0.1790	+0.4116	+0.4081	**0.4173**	0.0628	1.0539	0.6696	2.16
BRCA1	NG_005905.2	22/21	13.02x	+**0.1053**	−0.3336	−0.3336	−0.3545	+0.4388	+0.4597	**0.3598**	0.0544	0.9314	0.7019	5.30
PTEN	NG_007466.2	9/8	82.34x	+**0.1206**	−0.4099	−0.3819	−0.3882	+0.5025	+0.5088	**0.3781**	0.1323	1.2220	0.4329	6.23
RB1	NG_009009.1	27/26	62.21x	+**0.1055**	−0.3822	−0.3653	−0.3751	+0.4707	+0.4806	**0.3614**	0.0452	1.1083	0.3966	6.01
GAPDH	NG_007073.2	8/7	2.31x	+**0.2249**	−0.1513	−0.1513	−0.2674	+0.3761	+0.4923	**0.3713**	0.1985	1.2114	1.1769	3.09
ACTB	NG_007992.1	5/4	0.69x	+**0.2686**	−0.2826	−0.0207	−0.3150	+0.2892	+0.5835	**0.5401**	0.4385	1.1440	1.2762	3.85
HBB	NG_000007.3	3/2	2.21x	+**0.1911**	−0.3994	−0.2008	−0.4461	+0.3919	+0.6372	**0.4247**	0.2945	1.2120	1.2059	6.44
KRAS	NG_007524.2	4/3	51.53x	+**0.1454**	−0.4115	−0.3757	−0.4169	+0.5211	+0.5623	**0.4681**	0.0941	1.3055	0.6079	6.63
NRAS	NG_007572.1	4/3	12.38x	+**0.1004**	−0.3286	−0.3286	−0.3402	+0.4290	+0.4406	**0.4149**	0.1437	1.2964	0.9693	5.57
HRAS	NG_007666.1	4/3	1.96x	+**0.2448**	−0.0943	−0.0943	−0.1435	+0.3391	+0.3882	**0.5000**	0.1828	1.2635	1.1970	2.42
MYC	NG_007161.2	3/2	2.20x	+**0.1968**	−0.1906	−0.1850	−0.2522	+0.3818	+0.4490	**0.3996**	0.1700	1.1066	1.1220	3.50
APOE	NG_007084.2	3/2	1.75x	+**0.2681**	−0.1673	−0.1616	−0.1973	+0.4297	+0.4655	**0.4684**	0.3178	0.9597	1.1760	2.51
Mean	**14/14 loci**	—	—	**+0.1721**	**−0.2885**	**−0.2440**	**−0.3097**	**+0.4162**	**+0.4818**	**0.4176**	**0.1615**	**1.1578**	**0.8814**	**4.54**

Table 8: Autonomous four-state compartment audit across 14 human RefSeqGene loci (`v14.6`). In **14/14** genes ($p = 2^{-14} = 6.1 \times 10^{-5}$), conditional syntax beyond nucleotide trigrams (d_{3b}) is strictly positive in concatenated exons (with 0.00 internal stops/100 codons in every spliced CDS) and strictly negative in both whole unspliced pre-mRNA (across all three reading frames) and excised introns, while 32-pair mirror asymmetry (Mirror$_{L_1}$) is higher in spliced exons than in introns in **14/14** genes.

Gene	Structural state	Len (nt)	Codons	d_0	d_2	d_{2b}	d_3	d_{3b}	Δ_5	Δ_6	Δ_7	GC3	H_{codon}
TP53	2/ Exons concat (spliced)	1,176	392	+0.0935	+0.0734	+0.0671	+0.1314	+**0.1117**	+0.1208	+0.1447	+0.1324	0.6250	5.6085
	1/ Whole gene (frame 0)	6,984	2,328	−0.0179	−0.0239	+0.0081	−0.3108	−0.3176	−0.0544	+0.0947	+0.0757	0.5120	5.8432
	Whole gene (frame 1)	6,984	2,328	+0.0098	−0.0106	+0.0201	−0.2896	−0.2963	−0.0595	+0.1081	+0.0989	—	—
	Whole gene (frame 2)	6,984	2,328	+0.0006	−0.0200	+0.0185	−0.2488	−0.2554	−0.0370	+0.0859	+0.0881	—	—
	4/ Excised introns only	5,802	1,934	+0.0004	−0.0120	+0.0080	−0.2701	−0.2738	−0.0465	+0.0990	+0.0870	0.4938	5.8364
TP63	2/ Exons concat (spliced)	2,037	679	+0.0665	+0.0626	+0.0592	+0.1118	+**0.0976**	+0.0849	+0.1457	+0.1262	0.6097	5.6621
	1/ Whole gene (frame 0)	262,986	87,662	−0.0120	−0.0267	−0.0130	−0.3675	−0.3871	−0.1356	+0.0665	+0.0635	0.3727	5.7704
	4/ Excised introns only	260,943	86,981	−0.0094	−0.0256	−0.0116	−0.3668	−0.3865	−0.1356	+0.0672	+0.0662	0.3691	5.7674
TP73	2/ Exons concat (spliced)	1,905	635	+0.1072	+0.0935	+0.0376	+0.2446	+**0.2291**	+0.2056	+0.1144	+0.0985	0.8409	5.3497
	1/ Whole gene (frame 0)	50,712	16,904	+0.0359	+0.0227	+0.0437	−0.1767	−0.1826	+0.0545	+0.1229	+0.1115	0.6217	5.6716
	4/ Excised introns only	48,801	16,267	+0.0362	+0.0236	+0.0460	−0.1735	−0.1790	+0.0584	+0.1243	+0.1118	0.6217	5.6638
BRCA1	2/ Exons concat (spliced)	5,586	1,862	+0.0707	+0.0707	+0.0341	+0.1102	+**0.1053**	+0.0148	+0.1021	+0.0874	0.3588	5.5674
	1/ Whole gene (frame 0)	78,417	26,139	−0.0077	−0.0232	−0.0019	−0.3198	−0.3336	−0.1026	+0.0841	+0.0712	0.4235	5.8490
	4/ Excised introns only	72,825	24,275	−0.0113	−0.0324	−0.0089	−0.3400	−0.3545	−0.1065	+0.0804	+0.0694	0.4282	5.8441
KRAS	2/ Exons concat (spliced)	564	188	+0.0944	+0.0940	+0.0287	+0.1578	+**0.1454**	+0.0413	+0.0955	+0.0807	0.3298	5.3338
	1/ Whole gene (frame 0)	29,943	9,981	−0.0203	−0.0312	−0.0169	−0.3857	−0.4115	−0.1594	+0.0639	+0.0506	0.3557	5.7220
	4/ Excised introns only	29,373	9,791	−0.0192	−0.0309	−0.0173	−0.3908	−0.4169	−0.1614	+0.0625	+0.0488	0.3521	5.7210
NRAS	2/ Exons concat (spliced)	564	188	+0.0518	+0.0767	+0.0777	+0.1239	+**0.1004**	+0.0605	+0.1163	+0.1320	0.4574	5.3983
	1/ Whole gene (frame 0)	7,626	2,542	+0.0102	−0.0113	−0.0003	−0.3070	−0.3286	−0.1198	+0.0828	+0.0708	0.3788	5.7864
	4/ Excised introns only	7,056	2,352	+0.0047	−0.0111	−0.0025	−0.3186	−0.3402	−0.1161	+0.0833	+0.0713	0.4103	5.7845
HRAS	2/ Exons concat (spliced)	564	188	+0.0756	+0.0913	+0.0526	+0.2900	+**0.2448**	+0.2497	+0.1239	+0.1028	0.8138	5.1002
	1/ Whole gene (frame 0)	1,686	562	+0.0675	+0.0480	+0.0507	−0.0789	−0.0943	+0.1089	+0.1299	+0.1361	0.7206	5.5216
	4/ Excised introns only	1,116	372	+0.0772	+0.0694	+0.0530	−0.1430	−0.1435	+0.1124	+0.1247	+0.1358	0.7339	5.2832
HBB	2/ Exons concat (spliced)	438	146	+0.0822	+0.0838	+0.0791	+0.2064	+**0.1911**	+0.2605	+0.1271	+0.0940	0.6575	4.8268
	1/ Whole gene (frame 0)	1,422	474	−0.0103	−0.0277	−0.0200	−0.3634	−0.3994	−0.1344	+0.0772	+0.0372	0.3924	5.7008
	4/ Excised introns only	978	326	−0.0512	−0.0675	−0.0565	−0.4003	−0.4461	−0.2271	+0.0058	−0.0095	0.3344	5.5175

Table 9: Detailed multi-null breakdown for the $p53$ tumor-suppressor paralog family (*TP53*, *TP63*, *TP73*), the *RAS* proto-oncogene paralog family (*KRAS*, *NRAS*, *HRAS*), AT-rich *BRCA1*, and compact *HBB* (`v14.6`). Analytical residuals ($d_0, d_2, d_{2b}, d_3, d_{3b}$), GC3, and H_{codon} are deterministic point estimates; Δ_5, Δ_6, and Δ_7 are single-seed Monte Carlo surrogate/shuffle diagnostics (`mc_reps=40`).

whole-genome non-coding parity equilibrium of [6, 8] ($\text{Mirror}_{L_1} = 0.0951, 0.0311, 0.0628$; $d_{3b} = -0.2738, -0.3865, -0.1790$) to symmetry-broken exonic program ($\text{Mirror}_{L_1} = 0.3878, 0.3549, 0.4173$; $d_{3b} = +0.1117, +0.0976, +0.2291$; $\Delta_5 = +0.1208, +0.0849, +0.2056$). Even more strikingly, all three human *RAS* proto-oncogenes (*KRAS*, *NRAS*, *HRAS*) share an *identical* 4-exon / 3-intron coding architecture with exact exon-length conservation ($[111, 179, 160, 120]$ nt = 570 nt spliced CDS / 188 internal codons) and identical junction phases $[0, 2, 0]$, while diverging 26× in intronic length and spanning the full human GC3 spectrum from AT-rich *KRAS* ($\text{GC3}_{\text{ex}} = 0.3298$, $I/E = 51.53\times$) through intermediate *NRAS* ($\text{GC3}_{\text{ex}} = 0.4574$, $I/E = 12.38\times$) to extreme GC-rich *HRAS* ($\text{GC3}_{\text{ex}} = 0.8138$, $I/E = 1.96\times$). Across all three *RAS* paralogs (**3/3**), d_{3b} is strictly negative in introns ($-0.4169, -0.3402, -0.1435$) and unspliced pre-mRNA ($-0.3757, -0.3286, -0.0943$) and strictly positive in spliced exons ($+0.1454, +0.1004, +0.2448$; mean exon-minus-intron contrast +0.46370 nats/codon), while Mirror_{L_1} jumps from a mean of 0.1402 in introns to 0.4610 in exons (reaching 0.5000 in *HRAS*), and along all three 4-step progressive $5' \to 3'$ splicing trajectories d_{3b} remains negative through Step 2 (66.7% introns removed: $-0.3269, -0.1351, -0.0439$) before flipping positive at Step 3 (100% spliced).

3. **Invariance to GC3 gradient direction:** In GC-rich genes such as *TP53* ($\text{GC3}_{\text{ex}} = 0.6250$ vs. $\text{GC3}_{\text{in}} = 0.4938$), *TP73* (0.8409 vs. 0.6217), *HRAS* (0.8138 vs. 0.7339), and *APOE* (0.8987 vs. 0.5709), exons are more GC-rich than introns. Conversely, in AT-rich tumor suppressors and proto-oncogenes—*BRCA1* ($\text{GC3}_{\text{ex}} = 0.3588$ vs. $\text{GC3}_{\text{in}} = 0.4282$), *RB1* (0.3366 vs. 0.3694), and *KRAS* (0.3298 vs. 0.3521)—exons are *more AT-rich* than introns. Yet both groups exhibit the exact same sign inversion in d_{3b} (+0.4597 in *BRCA1*, +0.4806 in *RB1*, +0.5623 in *KRAS*, +0.3882 in *HRAS*) and the same jump in mirror asymmetry (Mirror_{L_1} rising 6.6× in *BRCA1*, 8.0× in *RB1*, 5.0× in *KRAS*, and 2.7× in *HRAS*).
4. **In-silico progressive $5' \to 3'$ splicing trajectory and local splice-junction reconstruction:** Table 10 reports both the 9-junction phase reconstruction audit and the 10-step progressive splicing trajectory ($5' \to 3'$) for *TP53* (`NG_017013.2`). Across the 9 canonical `GT..AG` junctions of *TP53*, spliced $\text{Exon}_k \to \text{Exon}_{k+1}$ codon pairs achieve a higher mean log-likelihood gain over H0 than unspliced $\text{Exon}_k \to \text{Intron}_k$ boundaries (+0.1782 vs. +0.1044 nats; mean advantage +0.0738 nats). Across the 14-gene panel, positive gene-mean junction advantages are observed in 10/14 loci—*TP53* (+0.0738), *PTEN* (+0.0571), *RB1* (+0.1067 across 26 junctions), *GAPDH* (+0.0179), *ACTB* (+0.2192), *HBB* (+0.2267), *KRAS* (+0.4295, 3/3 junctions positive), *NRAS* (+0.0598), *MYC* (+0.2189), and *APOE* (+0.0220)—whereas 4 loci exhibit slightly negative gene-mean advantages (*BRCA1* −0.0090 with 11/21 junctions positive, *TP63* −0.0156, *TP73* −0.0499, and compact high-GC *HRAS* −0.1479), reflecting strong local heterogeneity across individual splice sites (e.g. *RB1* junction 12 at +1.6141 and *BRCA1* junction 10 at +0.7094 vs. *BRCA1* junction 14 at −0.7700). Along the *TP53* in-silico progressive splicing trajectory, d_{3b} and Δ_5 remain negative from Step 0 (0% spliced, 6,986 nt span, $d_{3b} = -0.3176$) through Step 7 (77.8% spliced, $d_{3b} = -0.2573$), before excising the 2,819 nt intron 8 (Step 8, $\Delta_5 = +0.0649$, $\text{Mirror}_{L_1} = 0.2514$) and the final 918 nt intron 9 (Step 9, 100% spliced, 1,182 nt CDS span / 392 internal codons, $d_{3b} = +0.1117$, $\Delta_5 = +0.1258$, $\text{Mirror}_{L_1} = 0.3878$, $\text{FlowAsym}_F = 1.2353$, Stops/100 = 0.00).
5. **Exon-by-exon functional localization and single-codon oncogenic hotspot audit (*TP53* and *RAS* family; Table 11):** Partitioning spliced *TP53* across its 10 individual coding exons (`[74, 22, 279, 184, 113, 110, 137, 74, 107, 82]` nt) shows that 9 out of 10 exons (`E01`–`E09`, codons 1–366) carry strictly positive $d_{3b} > 0$, peaking in the central DNA-binding exons `E06` ($d_{3b} = +0.2394$, $d_{2b} = +0.1639$, $\text{Mirror}_{L_1} = 0.8889$), `E08` ($d_{3b} = +0.2011$), and `E05` ($d_{3b} = +0.1602$), whereas only the C-terminal basic regulatory tail in `E10` (codons 368–394) is negative ($d_{3b} = -0.1913$). Directly testing the regional hypothesis of Perez [9]—who noted that 86% of somatic *TP53* cancer mutations cluster between codons 125 and 300 (exons 5–8 DNA-binding domain)—reveals that codons 125–300 (176 codons) exhibit a **2.34**× higher d_{2b} (+0.0989 vs. +0.0423 nats/codon) and **1.78**× higher d_{3b} (+0.1454

vs. +0.0816 nats/codon) than the flanking coding regions (codons 2–124 and 301–393, 216 codons). Analogously, across all three 570-nt *RAS* proto-oncogenes (*KRAS*, *NRAS*, *HRAS*; 4 exons of `[111, 179, 160, 120]` nt [13]), the catalytic GTPase core in `E01`–`E03` (codons 1–150) is strictly positive on d_{3b} in **9/9** exons (*KRAS*: $+0.1479, +0.1965, +0.1439$; *NRAS*: $+0.1837, +0.0999, +0.0624$; *HRAS*: $+0.2572, +0.2998, +0.2347$), whereas the hypervariable C-terminal membrane-targeting tail in `E04` (codons 151–190) drops to negative or near-zero values (-0.1348 in *KRAS*, -0.1332 in *NRAS*, $+0.0237$ in *HRAS*). Evaluating single-codon oncogenic driver mutations across the 61 sense codons in the local two-transition window $c_{k-1} \to c_k \to c_{k+1}$ (Table 11b) uncovers two distinct Markovian mechanisms behind the mutational hotspots of [9, 13]: (i) *direct syntactic program collapse* ($\Delta d_{3b} < 0$), exemplified in *TP53* by `R175H` (`CGC->CAC`, Perez mutability coef 61/61, where local d_{3b} inverts sign from $+0.2793$ to -0.1413, $\Delta d_{3b} = -0.4207$ nats), `G245S` and `G245D` (`GGC->AGC/GAC`, coef 61/61, eroding an extreme wild-type program peak of $d_{3b} = +0.9940$ by -0.4214 and -0.1568 nats), `R248W` ($\Delta d_{3b} = -0.6074$ nats), `R273H` ($\Delta d_{3b} = -0.1531$ nats), and `R282W` (`CGG->TGG`, where local d_{3b} collapses from $+0.7587$ to -0.4531, $\Delta d_{3b} = -1.2117$ nats), and in the *RAS* family by transversion/melanoma drivers *KRAS* `G12C` ($\Delta\text{LL} = -0.7566$, flipping local d_{3b} from $+0.1328$ to -0.0471), *NRAS* `Q61R` ($\Delta\text{LL} = -1.3840$, $\Delta d_{3b} = -0.2912$), and *HRAS* `G12V` (Markov rank 46/61, $\Delta\text{LL} = -1.2580$, $\Delta d_{3b} = -0.2276$); and (ii) *statistical mutational-trap relaxation* ($\Delta\text{LL}_{\text{emp}} > 0$), where functionally constrained CpG arginine codons in *TP53* (`R248Q` $\Delta\text{LL} = +0.9483$; `R273H/C` Markov rank 44/61, $\Delta\text{LL} = +0.5591/ + 1.0526$) or tandem-glycine codons in *RAS* (*KRAS* `G13D` rank 37/61, $\Delta d_{3b} = +0.8689$; *HRAS* `G13D` rank 52/61, $\Delta d_{3b} = +0.3990$) are held in low-probability wild-type states that relax toward higher background transition probability upon spontaneous deamination.

The multi-null breakdown also clarifies the hierarchy of compartment markers. Because d_{3b} directly tests codon-bigram transitions against order-2 nucleotide trigrams across all reading frames, it serves as the primary compartment discriminator (14/14 sign separation), whereas amino-acid-conditioned metrics (d_{2b}, Δ_5) can be near zero in specific high-GC or constrained coding sequences (e.g. *APOE* exons $d_{2b} = -0.0143$ and *RB1* exons $d_{2b} = -0.0011$ despite strongly positive $d_{3b} = +0.2681$ and $+0.1055$). Similarly, *ACTB* provides an informative compact boundary case ($I/E = 0.69\times$): although its short, regulatory-rich introns already exhibit elevated mirror and flow asymmetry ($\text{Mirror}_{L_1} = 0.4385$, $\text{FlowAsym}_F = 1.2762$), its spliced exons still achieve a massive d_{3b} gain ($+0.2686$ vs. -0.3150, $\Delta d_{3b} = +0.5835$ nats/codon) and higher mirror asymmetry (0.5401). More generally, while mean Markov flow asymmetry (FlowAsym_F) is higher in exons than in introns across the panel (1.1578 vs. 0.8814), compact loci with short introns (*ACTB*, *MYC*, *APOE*) can exhibit intronic flow asymmetry comparable to or exceeding the exonic value; the primary 14/14 compartment signature is therefore the combined sign inversion of d_{3b}, reduced codon entropy, and increased mirror asymmetry in spliced exons.

6.6 Frontier biosphere and synthetic genome forensics audit: Giant viruses, deep subsurface extremophiles, Asgard archaea, and Venter's *JCVI-syn1.0* / *JCVI-syn3.0* (`v14.7`)

To test the predictive power of our ten-model hierarchy at the physical, evolutionary, and artificial boundaries of life, we extended the autonomous pipeline (`super_codon_proof_v14_6_7.py`, flag `--frontier-audit`) to 13 additional boundary genomes fetched directly from NCBI Entrez (`rettype=fasta_cds_na` and `gbwithparts`, evaluated with `family_split=False` so systematic `locus_tag` numbering does not collapse single-chromosome annotations):

1. **Suite 1 — Giant DNA viruses (*Nucleocytoviricota*):** *Acanthamoeba polyphaga mimivirus* (`NC_014649.1`, 1.18 Mb, 979 CDSs) [18], *Megavirus chilensis* (`NC_016072.1`, 1.26 Mb, 1,120 CDSs), *Pandoravirus salinus* (`NC_022098.1`, 2.47 Mb, 2,541 annotated CDSs / 1,430 deduplicated units with $\geq$ 20 transitions), and the 30,000-year Siberian permafrost giant virus

(a) TP53 *(NG_017013.2) Splice-Junction Phase Reconstruction Audit (mean advantage = +0.0738 nats):*

Junc	Exons	Phase	Intron (nt)	Sites	Spliced codon pair	Unspliced boundary	Spliced LL	Unspl. LL	Advantage
1	E01→E02	2	117	`GT..AG`	`AAA(K)->CTA(L)`	`AAA(K)->CTG(L)`	+0.2604	−0.5753	+0.8357
2	E02→E03	0	109	`GT..AG`	`CTG(L)->TCC(S)`	`CTG(L)->GTA(V)`	−0.0631	−0.1943	+0.1313
3	E03→E04	0	757	`GT..AG`	`ACG(T)->TAC(Y)`	`ACG(T)->GTC(V)`	−0.3477	+0.3525	−0.7002
4	E04→E05	1	81	`GT..AG`	`GAT(D)->GGT(G)`	`GAT(D)->GGT(G)`	+0.6060	+0.6060	+0.0000
5	E05→E06	0	568	`GT..AG`	`GAG(E)->GTT(V)`	`GAG(E)->GTC(V)`	−0.3660	−0.1682	−0.1978
6	E06→E07	2	343	`GT..AG`	`TCC(S)->AGT(S)`	`TCC(S)->AGG(R)`	+0.5594	+0.4331	+0.1263
7	E07→E08	1	92	`GT..AG`	`CGA(R)->GCA(A)`	`CGA(R)->GGT(G)`	+0.3276	+0.3998	−0.0722
8	E08→E09	0	2,819	`GT..AG`	`CAG(Q)->ATC(I)`	`CAG(Q)->GTA(V)`	−0.0329	−0.3473	+0.3143
9	E09→E10	2	918	`GT..AG`	`TCC(S)->AGC(S)`	`TCC(S)->AGG(R)`	+0.6602	+0.4331	+0.2271

(b) TP53 *In-silico 5′→3′ Progressive Splicing Trajectory (Steps 0 to 9):*

Step	Removed	Spliced %	Len (nt)	Codons	d_{2b}	d_{3b}	Δ_5	Δ_7	Mirror_{L_1} / FlowAsym_F (Stops)
0	0/9	0.0%	6,986	2,328	+0.0081	−0.3176	−0.0536	+0.0760	0.0962 / 1.0308 (4.12)
1	1/9	11.1%	6,869	2,289	+0.0068	−0.3193	−0.0550	+0.0787	0.0974 / 1.0373 (4.15)
2	2/9	22.2%	6,760	2,253	+0.0171	−0.2920	−0.0563	+0.0917	0.1203 / 1.0375 (4.04)
3	3/9	33.3%	6,003	2,001	+0.0146	−0.2225	−0.0414	+0.0890	0.1344 / 1.0434 (3.25)
4	4/9	44.4%	5,922	1,974	+0.0150	−0.2209	−0.0411	+0.0837	0.1378 / 1.0452 (3.24)
5	5/9	55.6%	5,354	1,784	+0.0065	−0.2788	−0.0510	+0.0775	0.1121 / 1.0851 (3.81)
6	6/9	66.7%	5,011	1,670	+0.0201	−0.2877	−0.0655	+0.0870	0.1437 / 1.1025 (4.13)
7	7/9	77.8%	4,919	1,639	+0.0034	−0.2573	−0.0555	+0.0730	0.1129 / 1.1014 (3.66)
8	8/9	88.9%	2,100	700	+0.0564	−0.0433	+0.0649	+0.1341	0.2514 / 1.1844 (1.57)
9	**9/9**	**100.0%**	**1,182**	**392**	**+0.0671**	**+0.1117**	**+0.1258**	**+0.1326**	**0.3878 / 1.2353 (0.00)**

Table 10: Splice-junction phase reconstruction and in-silico $5' \to 3'$ progressive splicing trajectory for human *TP53* (`NG_017013.2`, `v14.6`). In panel (b), `Len (nt)` reports the full structural span at each step, whereas `Codons` reports the scored triplet count after frame truncation and, at Step 9 (100% mature CDS, 1,182 nt), after start/stop codon trimming (1,176 nt = 392 internal codons).

(a) Exon-by-Exon Markov Profile of Human TP53 *(NG_017013.2) and the Three 570-nt* RAS *Paralogs (*KRAS, NRAS, HRAS*):*

Gene / Region	Exon / Span	Len (nt)	In-frame codons	Codon range	GC3	Mirror_{L_1}	d_0	d_{2b}	d_{3b}
TP53 (NG_017013.2)	E01..E04 (N-term / TAD / PRD)	559	185	1..186	0.6216	0.6432	+0.0706	+0.0365	+0.0841
	E05 (DBD hotspot exon)	113	37	188..224	0.4865	0.8378	+0.1567	+0.1286	**+0.1602**
	E06 (DBD peak exon)	110	36	225..260	0.8056	0.8889	+0.2164	+0.1639	**+0.2394**
	E07 (DBD hotspot exon)	137	45	262..306	0.5778	0.7333	+0.0374	+0.0298	**+0.0533**
	E08 (DBD hotspot exon)	74	24	308..331	0.6250	0.5833	+0.2193	+0.1543	**+0.2011**
	E09 (Tetramerization domain)	107	35	332..366	0.7429	0.7143	+0.0879	+0.0899	+0.1251
	E10 (C-terminal basic tail)	82	27	368..394	0.5926	0.6296	−0.0282	+0.0614	−0.1913
	Hotspot Core (Exons 5–8, 86% cancer muts)	**528**	**176**	**125..300**	**0.6761**	**0.4545**	**+0.1149**	**+0.0989**	**+0.1454**
	Flanking Coding Regions (Exons 1–4, 9–10)	**648**	**216**	**2..124, 301..393**	**0.5833**	**0.4630**	**+0.0759**	**+0.0423**	**+0.0816**
KRAS (NG_007524.2)	E01 / E02 / E03 (GTPase core)	111 / 179 / 160	37 / 59 / 53	1..150	0.3514 / 0.4068 / 0.2075	0.5676 / 0.4237 / 0.7358	+0.0028 / +0.1528 / +0.1043	+0.0078 / +0.0684 / +0.0061	**+0.1479** / **+0.1965** / **+0.1439**
	E04 (Hypervariable C-tail)	120	40	151..190	0.3750	0.7000	+0.0238	+0.0327	−0.1348
NRAS (NG_007572.1)	E01 / E02 / E03 (GTPase core)	111 / 179 / 160	37 / 59 / 53	1..150	0.4865 / 0.4576 / 0.4717	0.4054 / 0.4576 / 0.6604	+0.0765 / +0.0737 / −0.0106	+0.1435 / +0.0968 / +0.0329	**+0.1837** / **+0.0999** / **+0.0624**
	E04 (Hypervariable C-tail)	120	40	151..190	0.4000	0.6500	+0.0243	+0.0548	−0.1332
HRAS (NG_007666.1)	E01 / E02 / E03 (GTPase core)	111 / 179 / 160	37 / 59 / 53	1..150	0.7568 / 0.8644 / 0.8113	0.6216 / 0.6610 / 0.6981	−0.0167 / +0.1040 / +0.0579	+0.0048 / +0.0479 / +0.0456	**+0.2572** / **+0.2998** / **+0.2347**
	E04 (Hypervariable C-tail)	120	40	151..190	0.7750	0.6500	+0.1557	+0.1163	+0.0237

(b) Single-Codon Oncogenic Hotspot Mutation Markov Audit in TP53 *[9] and 570-nt* RAS *Paralogs [13]:*

Gene	Codon	Substitution (AA)	Local 3-codon window	Rank (1..61)	LL_{WT}	LL_{Mut}	$\Delta\text{LL}_{\text{emp}}$	$d_{3b,\text{WT}} \to d_{3b,\text{Mut}}$	Δd_{3b} (2-trans)
TP53	80	`CCT->CTT` (P80L, conserved ctrl)	`GCT-CCT->CTT-ACA`	31/61	−8.5095	−8.7844	−0.2749	−0.3499 → −0.4515	−0.1016
	175	`CGC->CAC` (R175H, Hotspot #1)	`AGG-CGC->CAC-TGC`	13/61	−7.8744	−7.9899	−0.1155	+0.2793 → −0.1413	**−0.4207**
	245	`GGC->AGC` (G245S, Hotspot #2)	`GGC-GGC->AGC-ATG`	13/61	−7.4330	−7.0590	+0.3740	+0.9940 → +0.5725	**−0.4214**
	245	`GGC->GAC` (G245D, Hotspot #2)	`GGC-GGC->GAC-ATG`	13/61	−7.4330	−8.1331	−0.7001	+0.9940 → +0.8371	−0.1568
	248	`CGG->CAG` (R248Q, Hotspot #3)	`AAC-CGG->CAG-AGG`	20/61	−8.2446	−7.2963	+0.9483	−0.3122 → −0.2191	+0.0931
	248	`CGG->TGG` (R248W, Hotspot #3)	`AAC-CGG->TGG-AGG`	20/61	−8.2446	−8.0639	+0.1807	−0.3122 → −0.9196	**−0.6074**
	273	`CGT->CAT` (R273H, Hotspot #4)	`GTG-CGT->CAT-GTT`	44/61	−9.6657	−9.1066	+0.5591	−0.3560 → −0.5092	−0.1531
	273	`CGT->TGT` (R273C, Hotspot #4)	`GTG-CGT->TGT-GTT`	44/61	−9.6657	−8.6131	+1.0526	−0.3560 → −0.2647	+0.0913
	282	`CGG->TGG` (R282W, Exon 8)	`GAC-CGG->TGG-CGC`	14/61	−8.0826	−8.5063	−0.4236	+0.7587 → −0.4531	**−1.2117**
KRAS	12	`GGT->GAT` (G12D, Exon 2 driver)	`GCT-GGT->GAT-GGC`	8/61	−7.1613	−7.0924	+0.0689	+0.1328 → +0.4073	+0.2745
	12	`GGT->GTT` (G12V, Exon 2 driver)	`GCT-GGT->GTT-GGC`	8/61	−7.1613	−7.5962	−0.4349	+0.1328 → +0.2992	+0.1664
	12	`GGT->TGT` (G12C, smoking driver)	`GCT-GGT->TGT-GGC`	8/61	−7.1613	−7.9179	**−0.7566**	+0.1328 → −0.0471	**−0.1799**
	13	`GGC->GAC` (G13D, Exon 2 driver)	`GGT-GGC->GAC-GTA`	37/61	−9.6523	−9.4077	+0.2445	−0.5529 → +0.3160	+0.8689
NRAS	61	`CAA->CGA` (Q61R, melanoma driver)	`GGA-CAA->CGA-GAA`	15/61	−7.1596	−8.5437	**−1.3840**	+0.4782 → +0.1870	**−0.2912**
HRAS	12	`GGC->GTC` (G12V, bladder driver)	`GCC-GGC->GTC-GGT`	46/61	−9.4906	−10.7486	**−1.2580**	−0.1071 → −0.3347	**−0.2276**
	13	`GGT->GAT` (G13D, Exon 2 driver)	`GGC-GGT->GAT-GTG`	52/61	−8.5528	−8.5153	+0.0375	−0.0581 → +0.3410	+0.3990

Table 11: Sub-genic exon-by-exon Markov profiles and single-codon oncogenic hotspot mutation audits (`v14.6`) for human *TP53* and the three 570-nt *RAS* proto-oncogenes (*KRAS*, *NRAS*, *HRAS*), extending the mutability and onco-harmonic frameworks of Perez [9, 13].

Pithovirus sibericum (NC_023423.1, 467 CDSs).

2. **Suite 2 — Deep subsurface "hell boreholes" and 122°C abyssal extremophiles:** *Candidatus Desulforudis audaxviator* MP104C (NC_010424.1, a single-species ecosystem isolated 2.8 km underground in South African gold-mine fracture water sustained by U/Th radiolysis in total absence of sunlight or oxygen), *Methanopyrus kandleri* AV19 (NC_003551.1, 2,000 m deep hydrothermal black smoker, holding the 122°C upper temperature record for life), *Aquifex aeolicus* VF5 (NC_000918.1, 95°C deep-branching chemolithoautotrophic bacterium), and *Pyrococcus furiosus* DSM 3638 (NC_003413.1, 100°C abyssal archaeon).
3. **Suite 3 — Venter's chemically synthesized cells vs. wild-type *Mycoplasma mycoides* (NCBI Table 4, TGA → W):** natural wild-type *Mycoplasma mycoides* subsp. *mycoides* SC PG1 (NC_005364.2), the 1.08 Mb chemically synthesized genome *JCVI-syn1.0* (CP002027.1) [19], and the 531 kb human-designed minimal cell *JCVI-syn3.0* (CP014940.1, 438 CDSs) [20].
4. **Suite 4 — Eukaryogenesis bridge and minimal cellular endosymbiont:** the cultured Asgard archaeon *Candidatus Prometheoarchaeum syntrophicum* strain MK-D1 (NZ_CP042905.1, closest known prokaryotic relative of eukaryotes) [21] and the 160 kb obligatory insect endosymbiont *Candidatus Carsonella ruddii* PV (NC_008512.1, 182 CDSs, smallest known non-organellar cellular genome).

(a) Within-Genome 10-Null Conditional Syntax Audit Across 13 Frontier Biosphere & Synthetic Genomes (v14.7):

Taxon / Construct	Suite	Accession	Tab	Trn/Tst	Test trans	GC3	Mirror$_{L_1}$	d_{2b}	d_{3b}	$\Delta_{H4}(30)$	Δ_5	Δ_6	Δ_7 (Verdict)
Mimivirus	1_giant_virus	NC_014649.1	1	783/196	68,196	0.1915	0.3462	+0.0178	+0.1770	+0.0019	+0.0382	+0.0580	+0.0530 (STRICT 7/7)
Megavirus_chil.	1_giant_virus	NC_016072.1	1	896/224	78,792	0.1564	0.3840	+0.0166	+0.2027	+0.0023	+0.0145	+0.0546	+0.0534 (STRICT 7/7)
Pandoravirus_sal.	1_giant_virus	NC_022098.1	1	1144/286	107,940	0.7549	0.2203	+0.0671	+0.1514	+0.0079	+0.1490	+0.2095	+0.1535 (STRICT 7/7)
Pithovirus_sib.	1_giant_virus	NC_023423.1	1	373/94	26,053	0.3677	0.2655	+0.0470	+0.1098	+0.0062	+0.0824	+0.1678	+0.1305 (STRICT 7/7)
Desulforudis_aud.	2_deep_borehole	NC_010424.1	11	1811/453	129,876	0.7675	0.2934	+0.0285	+0.2172	+0.0032	+0.0574	+0.0848	+0.0721 (STRICT 7/7)
Methanopyrus_kan.	2_deep_122C	NC_003551.1	11	1431/358	101,694	0.7644	0.2936	+0.0270	+0.1913	+0.0029	+0.0627	+0.0678	+0.0543 (STRICT 7/7)
Aquifex_aeolicus	2_deep_95C	NC_000918.1	11	1348/338	97,159	0.4830	0.3379	+0.0691	+0.2526	+0.0067	+0.1432	+0.2010	+0.1517 (STRICT 7/7)
Pyrococcus_fur.	2_deep_100C	NC_003413.1	11	1642/411	112,487	0.3943	0.3763	+0.0308	+0.1733	+0.0030	+0.0599	+0.0854	+0.0644 (STRICT 7/7)
M. mycoides_WT	3_natural_WT	NC_005364.2	4	840/211	67,935	0.1066	0.4445	+0.0208	+0.2153	+0.0029	+0.0329	+0.0901	+0.0735 (STRICT 7/7)
JCVI_syn1_0	3_synth_1.08Mb	CP002027.1	4	662/166	60,654	0.0887	0.4884	+0.0179	+0.2706	+0.0020	+0.0166	+0.0656	+0.0549 (6/7 POS CI)
JCVI_syn3_0	3_synth_531kb	CP014940.1	4	350/88	30,714	0.0752	0.5320	+0.0207	+0.2949	+0.0025	+0.0346	+0.0792	+0.0622 (STRICT 7/7)
Prometheoarch._MKD1	4_asgard_arch	NZ_CP042905.1	11	3131/783	253,357	0.2301	0.3452	+0.0114	+0.1430	+0.0010	+0.0253	+0.0318	+0.0292 (STRICT 7/7)
Carsonella_ruddii	4_minimal_cell	NC_008512.1	11	145/37	10,083	0.0757	0.3364	+0.0000	+0.1640	+0.0018	−0.0050	+0.0307	+0.0500 (4/7 POS CI)

(b) Blind Forensic Detection of Engineered Vector Genes & Intergenic Watermark Cassettes in Venter's Synthetic Genomes (Scored Under WT M. mycoides*):*

Construct	Locus / Span	Coordinates	Size	GC3 / GC	Mirror$_{L_1}$	d_0	d_{2b}	d_{3b} (f0)	d_{3b}^{best3f}	Forensic identification / annotation
JCVI_syn1_0	MMSYN1_0929	959417..959686	88 cod	0.4545	—	−0.2155	−0.0110	**−0.5294**	—	Engineered cassette ORF (lowest d_{3b} in *syn1.0*; 12/789 genes < 0)
(CP002027.1)	MMSYN1_0915	31535..34618	1026 cod	0.6335	—	−0.3184	−0.1812	**−0.4564**	—	**Foreign *E. coli* lacZ β-galactosidase** (blue-white screening marker)
	MMSYN1_0918	27639..28301	219 cod	0.4658	—	−0.4001	−0.1433	**−0.4286**	—	**Foreign yeast HIS3 dehydratase** (yeast centromeric vector marker)
	MMSYN1_0914	25426..26286	285 cod	0.4421	—	−0.2409	−0.1130	**−0.2353**	—	**Foreign bla TEM β-lactamase** (ampicillin resistance marker)
	IG_MMSYN1_0072	92028..97602	5575 nt	0.4254	0.2196	—	−0.0688	**−0.4626**	−0.4626	Synthetic watermark / vector cassette (4.57 stops/100 codons)
	IG_MMSYN1_0535	637181..642338	5158 nt	0.4344	0.2240	—	−0.0879	**−0.4825**	−0.4825	Synthetic watermark / vector cassette (3.49 stops/100 codons)
JCVI_syn3_0	**All 412 CDSs**	≥ 90 codons	412 gen	0.0752	0.5320	+0.0409	+0.0208	**+0.2792**	—	**0/412 retained essential *Mycoplasma* genes have** $d_{3b} < 0$ (min +0.0143)
(CP014940.1)	IG_JCVISYN3_0076	52290..58155	5866 nt	0.4133	0.2194	—	−0.0675	**−0.4559**	−0.4559	Synthetic intergenic cassette #1 (10/10 spans ≥ 600 nt negative in all 3 frames)
	IG_JCVISYN3_0025	18516..22768	4253 nt	0.3317	0.1588	—	−0.0940	**−0.4373**	−0.2424	Synthetic intergenic cassette #2 (4.66 stops/100 codons)
	IG_JCVISYN3_0606	358005..360644	2640 nt	0.3553	0.2000	—	−0.0645	**−0.4468**	−0.3457	Synthetic intergenic cassette #3 (3.86 stops/100 codons)

Table 12: Frontier biosphere and synthetic genome forensics audit (super_codon_proof_v14_6_7.py, v14.7). Panel (a) reports within-genome held-out metrics across Giant Viruses, Deep Subsurface Extremophiles, Venter's Synthetic Cells, Asgard Archaea, and Minimal Endosymbionts (with a 25,000-transition guard for compact genomes). Panel (b) reports blind forensic scoring of *JCVI-syn1.0* and *JCVI-syn3.0* against the wild-type *Mycoplasma mycoides* transition model (Table 4).

Table 12 reveals four striking results:

1. **Giant DNA viruses carry a full cellular-grade syntactic program (4/4 STRICT 7/7):** Unlike the 11-ORF RNA virus SARS-CoV-2 (Table 2), all four giant DNA viruses—*Mimivirus*, *Megavirus chilensis*, *Pandoravirus salinus*, and 30,000-year permafrost *Pithovirus sibericum*—pass all seven strict bootstrap criteria across a 60 percentage-point GC3 span (0.1564 to 0.7549), with *Pandoravirus salinus* ($d_{2b} = +0.0671$, $d_{3b} = +0.1514$, $\Delta_5 = +0.1490$, $\Delta_6 = +0.2095$, $\Delta_7 = +0.1535$) exceeding most free-living eukaryotes in conditional syntax strength. Moreover, giant viruses preserve a more complete conditional syntax than the 160 kb minimal cellular endosymbiont *Candidatus Carsonella ruddii* (NC_008512.1, GC3 = 0.0757, 182 CDSs), where extreme AT-driven reductive evolution erodes synonymous choice ($d_{2b} = 0.0000$, $\Delta_5 = -0.0050$, 4/7 POS CI) while leaving only the nucleotide-trigram and mirror/synonymous shuffle residuals ($d_{3b} = +0.1640$, $\Delta_6 = +0.0307$, $\Delta_7 = +0.0500$).
2. **Universal 30-codon syntax in isolated 2.8 km deep boreholes and at 122°C (4/4 STRICT 7/7):** All four deep-biosphere extremophiles pass STRICT 7/7, confirming that the ∼ 30-codon sequential invariant holds in a single-species radiolytic ecosystem isolated 2.8 km

underground (*Desulforudis audaxviator*, $d_{2b} = +0.0285$, $d_{3b} = +0.2172$), at the 122°C upper temperature limit of life (*Methanopyrus kandleri*, $d_{2b} = +0.0270$, $d_{3b} = +0.1913$), and at the prokaryotic bridge to eukaryotes in the Asgard archaeon *Prometheoarchaeum syntrophicum* MK-D1 ($d_{2b} = +0.0114$, $d_{3b} = +0.1430$, 253,357 test transitions).

3. **Blind forensic detection of human-engineered foreign vector genes in *JCVI-syn1.0*:** When Venter's first chemically synthesized cell *JCVI-syn1.0* (`CP002027.1`) is evaluated within-species, its H5 surrogate excess drops by half relative to wild-type *Mycoplasma mycoides* (+0.0166 vs. +0.0329, `6/7 POS CI`). Scoring all 789 genes ($\geq$ 90 codons) of *JCVI-syn1.0* under the wild-type *M. mycoides* Markov model (Table 12b) explains why: without reading any gene annotations, our algorithm flags exactly **12/789** loci with negative $d_{3b} < 0$, and those top outliers are the exact foreign engineering markers inserted by the JCVI team during yeast/bacterial assembly [19]—specifically *E. coli* β-galactosidase `lacZ` (`MMSYN1_0915`, 1,026 codons, $d_{2b} = -0.1812$, $d_{3b} = -0.4564$), yeast `HIS3` imidazoleglycerol-phosphate dehydratase (`MMSYN1_0918`, $d_{2b} = -0.1433$, $d_{3b} = -0.4286$), plasmid `bla` TEM β-lactamase (`MMSYN1_0914`, $d_{2b} = -0.1130$, $d_{3b} = -0.2353$), and flanking synthetic cassette ORFs (`MMSYN1_0929`, $d_{3b} = -0.5294$; `MMSYN1_0928`, -0.2672).
4. **Complete preservation of native coding program vs. intergenic watermark collapse in *JCVI-syn3.0*:** In the reduced minimal cell *JCVI-syn3.0* (`CP014940.1`) [20], where the foreign `lacZ`/`HIS3`/`bla` selection markers were removed and 412 essential *Mycoplasma* genes ($\geq$ 90 codons) were retained, **0/412 protein-coding genes** exhibit $d_{3b} < 0$ (whole-proteome cross-score $d_{2b} = +0.0208$, $d_{3b} = +0.2792$; within-species `STRICT 7/7` with $\text{Mirror}_{L_1} = 0.5320$). Conversely, across all **10/10** large intergenic non-coding cassettes ($\geq$ 600 nt) in *JCVI-syn3.0* and the synthetic watermark cassettes in *JCVI-syn1.0* (which house ASCII-encoded author names and literary quotes [19]), d_{3b} collapses to strongly negative values across all three reading frames ($d_{3b,\text{f0}} \in [-0.4825, -0.1964]$, $d_{3b}^{\text{best3f}} \in [-0.4825, -0.1378]$, $\text{Mirror}_{L_1} \in [0.1588, 0.3162]$). Thus, the Markovian syntax audit cleanly separates natural protein-coding programs from both foreign transgenic insertions and human-written ASCII DNA watermarks.
5. **Unifying macroscopic "form" disharmony (Perez 2022) with microscopic "program" forensics across synthetic life:** These results directly complement and explain the macroscopic whole-genome findings of Perez [11] on artificial genomes. At the whole-chromosome scale (*form* / topological epigenetics), Perez [11] proved that chemical rewriting and deletion of transposons/repeats destroy global palindromic forward-versus-backward Fibonacci ($3 \ldots 89$) and Lucas ($3 \ldots 76$) `TC/AG` and `TG/AC` resonance balance—inflating genome-weighted forward/backward resonance distance by 14.39× in synthetic *Caulobacter ethensis C. eth-2.0* (`CP035535.1` vs. native *NA1000* `NC_011916.1`), and driving a monotonic divergence of the Fibonacci-89 `TC/AG` forward/backward ratio across the 6-genome *Mycoplasma mycoides* series from natural wild-type (1.1328 in `NC_005364.2` and 1.1456 in `CP010267.1`) to the transgenic `tetM-lacZ` strain (`CP001621.1`, 1.1950), *JCVI-syn1.0* (1.2023), *JCVI-syn3.0* (1.2131), and *JCVI-syn3A* (`CP016816.2`, 1.2798), with analogous breakdowns in 61-codon recoded *E. coli Syn61* ($0.9704 \rightarrow 0.8665$ at Fibonacci 89) and synthetic yeast *Sc2.0* Chromosome XII `PCRTags` ($0.9976 \rightarrow 0.8482$ at Lucas 29 and $1.0894 \rightarrow 0.8607$ at Fibonacci 89) [11]. In Table 12a, our coding-compartment sense-strand mirror asymmetry Mirror_{L_1} tracks the exact same monotonic progression across the Venter series (0.4445 in wild-type `NC_005364.2` $\rightarrow$ 0.4884 in *JCVI-syn1.0* $\rightarrow$ 0.5320 in *JCVI-syn3.0*). Crucially, our Markovian decomposition resolves how *JCVI-syn3.0* remains biologically viable despite losing global chromosome parity: inside all **412 retained essential protein-coding genes**, the local 29-transition ($L = 30$ codon) syntactic *program* is 100% preserved ($d_{3b} > 0$), whereas the syntactic collapse is localized to the foreign `lacZ`/`HIS3`/`bla` vector insertions (the exact same `lacZ` marker class present in the transgenic `CP001621.1` strain of [11]) and non-coding ASCII watermarks, while synonymous recoding of intra-ORF `PCRTags` (*synXII*) or serine codons (*Syn61*) directly perturbs the empirical codon-pair transition matrix $P(c_{i+1} \mid c_i)$ measured by our d_{2b}, d_{3b}, Δ_5, and Δ_6 controls.

6.7 Viral "Genetic Trojan" specificity, endosymbiotic organelle rewriting, and the biotech "Max-CAI" codon-optimization crash test (v14.8)

To test whether conditional Markovian syntax governs host–pathogen compatibility, endosymbiotic gene assimilation, and synthetic gene design in biotechnology, we implemented a four-module audit in `super_codon_proof_v14_8.py` (flag `--chimera-audit`, Table 13):

(a) Module D — Biotech "Max-CAI Codon Optimization" Crash Test Across 14 Human RefSeqGene Loci (100% Identical Protein Sequences):

Gene	Accession	Codons	WT d_{2b}	MaxCAI d_{2b}	Δd_{2b}	WT d_{3b}	H0_Syn d_{3b}	Markov_H2b d_{3b}	WT H_{cod} (b)	MaxCAI H_{cod} (b) (Verdict)
TP53	NG_017013.2	392	+0.0677	**−0.0340**	−0.1017	+0.1110	+0.0436	+0.0736	5.6085	4.1227 (MAX-CAI CRASH)
TP63	NG_007550.1	679	+0.0599	**−0.0220**	−0.0819	+0.0991	+0.0467	+0.0863	5.6621	4.1744 (MAX-CAI CRASH)
TP73	NG_017035.2	635	+0.0380	**−0.0494**	−0.0874	+0.2313	+0.0210	+0.0968	5.3497	4.1577 (MAX-CAI CRASH)
BRCA1	NG_005905.2	1862	+0.0337	**−0.0478**	−0.0815	+0.1026	+0.0684	+0.1118	5.5674	4.0837 (MAX-CAI CRASH)
PTEN	NG_007466.2	402	+0.0280	**−0.0418**	−0.0698	+0.1172	+0.1161	+0.1127	5.5320	4.1891 (MAX-CAI CRASH)
RB1	NG_009009.1	927	−0.0006	**−0.0466**	−0.0460	+0.1053	+0.0721	+0.1228	5.6121	4.1349 (MAX-CAI CRASH)
KRAS	NG_007524.2	188	+0.0279	**−0.0447**	−0.0726	+0.1435	+0.0978	+0.1292	5.3338	4.1025 (MAX-CAI CRASH)
NRAS	NG_007572.1	188	+0.0787	**−0.0658**	−0.1445	+0.1013	+0.1343	+0.1340	5.3983	4.1214 (MAX-CAI CRASH)
HRAS	NG_007666.1	188	+0.0511	**−0.0763**	−0.1274	+0.2447	+0.0607	+0.1464	5.1002	4.1281 (MAX-CAI CRASH)
GAPDH	NG_007073.2	334	+0.1168	**−0.0904**	−0.2072	+0.2258	+0.1072	+0.1227	5.2701	4.1029 (MAX-CAI CRASH)
ACTB	NG_007992.1	374	+0.0134	**−0.0754**	−0.0888	+0.2699	+0.0329	+0.1143	5.1021	4.1835 (MAX-CAI CRASH)
HBB	NG_000007.3	146	+0.0814	**−0.1138**	−0.1952	+0.1958	+0.0513	+0.1415	4.8268	3.9313 (MAX-CAI CRASH)
MYC	NG_007161.2	453	+0.0164	**−0.0612**	−0.0776	+0.1991	+0.0687	+0.1202	5.4529	4.0533 (MAX-CAI CRASH)
APOE	NG_007084.2	316	−0.0139	**−0.0559**	−0.0420	+0.2681	+0.0371	+0.0686	4.6922	3.7820 (MAX-CAI CRASH)
Mean	**14/14 loci**	—	**+0.0427**	**−0.0589**	**−0.1016**	**+0.1725**	**+0.0684**	**+0.1129**	**5.3220**	**4.0905** (**14/14** $d_{2b} < 0$)

(b) Module A — Viral "Genetic Trojan" & Host-Ribosome Specificity Audit (Human_T2T OS vs. E. coli *OS):*

Virus / Construct	Accession	Genes	GC3	d_{2b}^{Hum}	d_{3b}^{Hum}	d_{2b}^{Eco}	d_{3b}^{Eco}	Pos_{Hum}	Top "Genetic Trojan" Host-Mimic Loci on Human Ribosome
EBV (HHV-4)	NC_007605.1	84	0.6819	+0.0183	**+0.0989**	−0.0699	−0.0791	76/82	BLLF3 dUTPase ($d_{2b} = +0.0538$), BBRF3 gM (+0.0519), BaRF1 RNR ($+0.0517, d_{3b} = +0.1960$)
Adenovirus C2	AC_000008.1	36	0.6476	−0.0127	**+0.1105**	−0.0377	−0.0377	33/33	E3 RID-α stealth (**$\mathbf{d_{2b} = +0.0958}$**), hexon capsid ($+0.0270, d_{3b} = +0.1468$), pVII (+0.2309)
HPV-16 (oncogenic)	NC_001526.4	8	0.2534	−0.0266	**+0.0970**	−0.0366	−0.0952	7/8	**E7 oncoprotein** is the **sole gene with** $d_{2b} > 0$ (**$+0.0327$**, $d_{3b} = +0.0682$); L1/L2 negative
Vaccinia poxvirus	NC_006998.1	215	0.3000	−0.0714	**+0.1145**	−0.0465	−0.0623	180/185	B7R ($d_{3b} = +0.2335$), A9L (+0.2195), A53R TNF-receptor mimic ($d_{2b} = -0.0050, d_{3b} = +0.1470$)
Bacteriophage T4	NC_000866.4	254	0.2422	−0.0392	+0.1271	−0.0260	**+0.0183**	197/200	Bacterial control: only virus with $d_{3b} > 0$ on *E. coli* host OS (+0.0183); pseT.3 $d_{2b}^{\text{Eco}} = +0.0114$

(c) Modules B & C — Endosymbiotic Organelle Rewriting & Retroviral/Transposon Assimilation Clock Inside the Human Genome:

Locus / Organelle	Accession	Size	GC3	d_{2b}^{Hum}	d_{3b}^{Hum}	Alt OS d_{3b} Contrast	Evolutionary Assimilation / Compartment Verdict
Arabidopsis Plastid	NC_000932.1	67 gen	0.2763	—	—	+0.0462 (Plant) vs. −0.0419 (Cyano)	**65/67 plastid genes** (97.0%, $p = 3.9 \times 10^{-17}$) **rewritten to host plant nuclear OS**
Human Mitochondrion	NC_012920.1	13 gen	0.4769	−0.0152	−0.1175	−0.1175 (Hum) vs. −0.1480 (Bact)	**10/13 mtDNA genes** (76.9%) **closer to human nuclear OS than bacterial ancestor**
CS (Citrate Synth.)	NM_004077.3	465 cod	0.5484	+0.0838	**+0.0924**	Zebrafish (+0.0841)	Ancient α-proteobacterial import: **100% assimilated** into human nuclear syntax
RAG1 Recombinase	NM_000448.3	1042 cod	0.5576	+0.0724	**+0.0837**	Zebrafish (+0.0674)	Ancient *Transib* transposase (∼ 500 Myr): **100% assimilated** into vertebrate syntax
ERVW-1 (Syncytin-1)	NM_014590.4	537 cod	0.4209	+0.0288	**+0.0304**	Arabidopsis (+0.0167)	Younger primate retroviral envelope (∼ 25–40 Myr): **intermediate assimilation** ($3.8\times$ **lower** d_{3b})
ARC Neuronal Capsid	NM_015193.5	395 cod	0.9595	−0.0090	+0.3186	*H. salinarum* (+0.3010)	Ty3/gypsy retrotransposon Gag capsid: extreme GC3 (0.9595), negative synonymous $d_{2b} = -0.0090$

Table 13: Translational and evolutionary forensics audit (`super_codon_proof_v14_8.py`, `v14.8`). Panel (a) contrasts wild-type human coding sequences against standard industrial `Max-CAI` codon optimization, memoryless `H0` synonymous sampling, and first-order `Markov_H2b` compilation. Panel (b) evaluates viral host-OS specificity and blind "Genetic Trojan" effector detection. Panel (c) quantifies endosymbiotic organellar rewriting and the assimilation clock of domesticated retroviral/transposon genes.

1. **The biotech "Max-CAI codon-optimization" crash across 14/14 human genes ($p = 2^{-14} = 6.1 \times 10^{-5}$, Table 13a):** Standard industrial codon-optimization algorithms used in synthetic biology, gene therapy, and mRNA vaccine/therapeutic design maximize the Codon Adaptation Index (`Max-CAI`), replacing every amino acid with the single most frequent synonymous codon in the human genome (zero-order marginal form H0/H2). When we apply `Max-CAI` to the 14 benchmark human RefSeqGene coding sequences (preserving 100% of the amino-acid sequence), the conditional codon-pair residual d_{2b} crashes from a wild-type mean of +0.0427 nats/codon into **strictly negative values in 14 out of 14 genes** (panel mean −0.0589 nats/codon; mean collapse $\Delta d_{2b} = -0.1016$ nats/codon), while 64-codon Shannon entropy drops by 1.23 bits (5.3220 → 4.0905 bits). The most severe collapses occur in high-expression housekeeping and globin genes such as *GAPDH* (+0.1168 → −0.0904) and *HBB* (+0.0814 → −0.1138), as well as *NRAS* (+0.0787 → −0.0658) and *TP53* (+0.0677 → −0.0340). Physically, greedy single-codon maximization creates repetitive, sterically incompatible adjacent tRNA pairs in the ribosomal A and P sites that natural coding sequences actively avoid ($d_{2b} < 0$), while artificially inflating raw nucleotide-trigram periodicity ($d_{3b} = +0.2855$). Conversely, memoryless synonymous sampling from human marginal frequencies (`H0_Syn`) loses 60.3% of wild-type d_{3b} (+0.1725 → +0.0684), whereas our first-order conditional Markov compiler (`Markov_H2b`) restores d_{3b} to +0.1129 nats/codon while preserving natural codon-pair transitions.
2. **Viral "Genetic Trojan" specificity and blind identification of oncogenic and stealth effectors (Table 13b):** Scoring whole viral proteomes simultaneously against the human (`Human_T2T`) and bacterial (*E. coli*) 64 × 64 Markov transition matrices cleanly separates host operating systems: all four human DNA viruses—Epstein–Barr virus (`NC_007605.1`), Human Adenovirus C2 (`AC_000008.1`), Human Papillomavirus 16 (`NC_001526.4`), and Vaccinia (`NC_006998.1`)—achieve strictly positive d_{3b} on the human OS (+0.0970 to +0.1145, with

87.5%–100% of individual viral genes positive) and strictly negative d_{3b} on *E. coli* (-0.0377 to -0.0952), whereas bacteriophage T4 (`NC_000866.4`) is the sole virus achieving positive $d_{3b} = +0.0183$ on its native *E. coli* host. Even more remarkably, blind per-gene scoring inside human viruses pinpoints the exact molecular "Trojan horses" responsible for host takeover:

- In high-risk oncogenic **HPV-16**, 7 out of 8 genes (including capsid proteins `L1` and `L2`) maintain negative synonymous $d_{2b} < 0$ on the human matrix (-0.0210 to -0.0312) to throttle translation and evade immune surveillance, whereas the **single gene in the entire HPV-16 genome with positive human synonymous syntax ($d_{2b} = +0.0327, d_{3b} = +0.0682$) is the transforming oncoprotein** `E7`, which degrades human retinoblastoma (*RB1*) protein to drive malignant proliferation.
- In **Human Adenovirus C2**, the #1 highest-scoring gene on the human synonymous matrix is `E3 RID-`α (`AP_000222.1`, $d_{2b} = +0.0958$, nearly double the native human genome mean of $+0.0495$)—the viral stealth protein that internalizes and degrades host Fas and TRAIL death receptors to block cytotoxic T-cell apoptosis—followed by the major capsid protein `hexon` (`AP_000211.1`, $d_{2b} = +0.0270, d_{3b} = +0.1468$).
- In **Epstein–Barr virus (HHV-4)**, the top human-syntax mimics are the host-pirated nucleotide-biosynthesis enzymes `BLLF3` **dUTPase** ($d_{2b} = +0.0538, d_{3b} = +0.1537$) and `BaRF1` **ribonucleotide reductase subunit 2** ($d_{2b} = +0.0517, d_{3b} = +0.1960$), alongside the DNA replication core `BALF5`/`BALF2`.

3. **Endosymbiotic organelle OS rewriting and the vertebrate syntactic firewall (Table 13c):** Evaluating the 67 protein-coding genes (≥ 40 codons) of the *Arabidopsis thaliana* chloroplast genome (`NC_000932.1`) reveals that the plastid proteome now scores negatively under its ancestral free-living cyanobacterial OS (*Synechocystis* sp. PCC 6803, $d_{3b} = -0.0419$) but positively under its host plant nuclear OS (*Arabidopsis*, $d_{3b} = +0.0462$), with **65 out of 67 chloroplast genes (**97.0%**,** $p = 3.9 \times 10^{-17}$**)** assimilated to the host plant syntax and only 2 genes retaining cyanobacterial preference (similarly, **10/13** human mitochondrial genes score closer to `Human_T2T` than to *B. subtilis*). Across 1,500 nuclear human genes (≥ 150 codons) scanned against 13 multi-kingdom OS kernels, 93.1% (1,396/1,500 on d_{2b}) and 94.4% (1,416/1,500 on d_{3b}) are won by vertebrate matrices, with 0/1,500 genes simultaneously beating the human model on a non-vertebrate OS by $\Delta d_{2b} > 0.035$ and $\Delta d_{3b} > 0.050$. Instead of retaining raw foreign syntax, horizontal and endosymbiotic acquisitions obey a **syntactic assimilation clock**: ancient imports such as mitochondrial citrate synthase (*CS*, $d_{2b} = +0.0838, d_{3b} = +0.0924$) and the *Transib*-derived immune recombinase (*RAG1*, $d_{2b} = +0.0724, d_{3b} = +0.0837$) are completely rewritten to native human syntax, whereas the younger HERV-W placental fusogen *ERVW-1/Syncytin-1* (`NM_014590.4`, integrated $\sim$ 25–40 Myr ago) exhibits a 3.8× lower d_{3b} ($+0.0304, d_{2b} = +0.0288$), and the hyper-GC neuronal Ty3/gypsy capsid *ARC* (`NM_015193.5`, GC3 = 0.9595) retains negative synonymous $d_{2b} = -0.0090$.

7 Discussion and Limitations

7.1 Synthesis: coexistence of static parity form and branching dynamic syntax

Our results provide an information-theoretic resolution to how genome-wide parity rules and clade-specific coding grammars coexist:

- **Static form is absorbed by H0, H1, H3, H3b, and H7 (and dominates in introns):** Marginal codon frequencies, dinucleotide/trigram biases, and the 32 reverse-complement codon-mirror pair populations [15, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14] constrain the *inventory* of codons in a sequence. In split eukaryotic genes (Section 6.5), excised introns and whole unspliced pre-mRNAs reside in this high-entropy, mirror-symmetric regime ($\text{Mirror}_{L_1} \approx 0.03$–$0.13$ in large multi-intron loci) and carry negative d_{3b} across all reading frames.
- **Dynamic program is measured by the surviving exonic residual:** Across all 52 guard-

passing cellular species (with the 2 additional small-split assemblies likewise showing strictly positive point estimates and 95% CIs across all seven criteria), the empirical transition matrix $P(c_{i+1} \mid c_i)$ extracts significant sequential information at $L = 30$ codons beyond the ten-model null hierarchy. These compartment data support an operational compression-operator interpretation of splicing: excising introns and concatenating exons shifts the sequence from a relatively higher-entropy, more mirror-symmetric intronic compartment to a symmetry-broken (mean $\text{Mirror}_{L_1} = 0.4176$), more time-directed (mean $\text{FlowAsym}_F = 1.1578$) coding compartment with $d_{3b} > 0$ in 14/14 human RefSeqGene loci. Across species, this dynamical layer is conserved at the coarse scale across examined jawed vertebrates, and conserved at the fine synonymous-choice scale within mammalian orders with a directional Primate → Rodent inclusion hierarchy.

Because the H5 conservation metric is normalized against the target species' within-species H5 baseline ($\Delta_5{}_T^{\text{within}}$), differences in baseline H5 strength contribute to directional gap asymmetry. Rodents in our dataset have elevated within-species H5 deltas (+0.1134 to +0.1143) relative to primates (+0.0969 to +0.1043). Consequently, primate-derived surrogates scored on rodent targets nearly match the rodent baseline (small negative gaps), whereas rodent-derived surrogates scored on primate targets produce higher empirical–surrogate divergence than native primate surrogates (positive overshoot). The key biological finding is that this baseline and transfer separation occurs at very small GC3 distances (ΔGC3 = 0.000–0.011 across the core and focal mammal contrasts; 0.000–0.008 within the 12-mammal matrix) and minimal acidic-proteome distances ($|\Delta\text{acid}| \leq 0.002$ across all primate–rodent pairs), and is replicated across all examined primate and rodent species, consistent with stronger purifying selection on local codon-pair preferences in Murinae (higher effective population size N_e) superimposed on a shared Euarchontoglires backbone.

7.2 Methodological limitations

1. **Borderline H5-transfer classification and point-estimate transfer metrics:** While all within-species deltas carry formal 95% bootstrap CIs, cross-species transfer deltas (d_{2b}^{transfer}) and H5-transfer gaps are point estimates evaluated against pre-specified thresholds. Because the within-species H5 bootstrap machinery already exists in `h5_bootstrap()`, extending it to source-generated cross-species surrogates is a straightforward next-release upgrade rather than a conceptual obstacle. Borderline pairs near $|\text{gap}| = 0.005$ (such as Human/Chimp → Mouse) can shift between `H5-CONSERVED` and `H5-NEAR-MISS` across runs; borderline calls should therefore be read as sensitivity-tier evidence rather than rigid binary boundaries.
2. **Phylogenetic non-independence and multiple comparisons:** Directed species pairs within mammals, vertebrates, and other clades share evolutionary history and are not statistically independent, and no family-wise error correction is applied across transfer matrices. Pairwise transfer counts are descriptive clade-level summaries rather than independent hypothesis tests.
3. **Orthogroup versus heuristic family splitting and gene-level bootstrap:** While one-gene-one-vote deduplication and `family_root()` prefix grouping reduce isoform leakage, divergent paralogs with distinct gene symbols can still span train/test splits, which could make gene-level bootstrap CIs mildly anti-conservative. Conversely, in assemblies annotated predominantly with systematic `locus_tag` prefixes (such as *Leishmania major* and *Staphylococcus aureus*), suffix stripping groups most loci into a single training cluster, leaving fewer than 50,000 test transitions and triggering destination-side small-sample skips. Future versions should implement sequence-clustering orthogroups (e.g. MMseqs2/OrthoFinder) and block-bootstrap resampling by `family_root()` cluster rather than individual gene; this would make the confidence intervals conservative with respect to residual gene-family correlation.
4. **Implicit NCBI nuclear translation table tags in *Candida albicans*:** Although `v14.5` implements and unit-tests exact Table 12 (`CTG` → `S`), NCBI RefSeq nuclear FASTA headers for *Candida albicans* omit explicit `[transl_table=12]` tags on individual records, causing the

header parser to default to Table 1. While *C. albicans* easily passes within-species `STRICT`, *Candida*-specific fine-transfer comparisons in `ext_fungi` should be treated as preliminary pending a species-level Table 12 metadata override.

5. **Absence of expression covariates and dN/dS selection tests:** This pipeline operates purely on RefSeq CDS and RefSeqGene sequences. Stratifying $\Delta_{H4}(30)$ and Δ_5 by empirical translation efficiency (Ribo-seq), mRNA secondary structure (DMS-seq), and ortholog dN/dS ratios is required to separate mutational and structural mechanisms from adaptive translational selection.
6. **Compartment audit reference-model overlap and per-locus inference (`v14.6`):** In the 14-gene RefSeqGene compartment audit, the intronic baseline and all sequence-level geometric invariants (Mirror_{L_1}, H_{codon}, FlowAsym_F, stop density) are completely free of RefSeq CDS training-set overlap: introns are absent from CDS training files, and the geometric invariants depend solely on the locus sequence itself. Because the reference transition matrix is pooled across $\sim$ 4,800 human `T2T-CHM13v2.0` training CDSs ($\sim 3.6 \times 10^6$ transitions from `max_records=6000`), any single target gene contributes $< 10^{-3}$ of total reference transition counts; nevertheless, the presence of a target exon or its paralogs in the training split can in principle impart a slight upward bias to absolute exonic residuals (and thus to the exact magnitude of Δd_{3b}) until per-family held-out reference models are applied. Furthermore, each locus is evaluated as a single compartmental unit rather than a multi-gene population; accordingly, cross-locus inference relies on the 14/14 nonparametric sign consistency ($p = 2^{-14} = 6.1 \times 10^{-5}$) and deterministic analytical residuals (d_0–d_{3b}), whereas single-locus Δ_5–Δ_7 values (`mc_reps=40`) and in-silico $5' \to 3'$ intron-removal trajectories serve as auxiliary diagnostics rather than kinetic models of spliceosome assembly. Finally, concatenated introns and unspliced pre-mRNAs are evaluated in fixed computational reading frames (frames 0, 1, and 2) strictly as a matched null-model control; no biological translation of introns is implied, and per-family held-out reference models will further refine absolute exonic effect sizes in future releases.

8 Conclusion

Across 54 cellular RefSeq genomes spanning the major branches of life and the sampled biological GC3 spectrum (0.1416 to 0.9177), real coding sequences in all 52 guard-passing cellular species pass the full `STRICT` stack at $L = 30$ codons (and the 2 remaining small-split cellular assemblies likewise exhibit strictly positive point estimates and 95% bootstrap CIs across all seven criteria), demonstrating a medium-range ($\sim$ 30 codons) conditional codon syntax $P(c_{i+1} \mid c_i)$ that survives the ten-model null hierarchy (including order-2 nucleotide trigrams H3b, amino-acid-preserving codon-pair surrogates H5, gene-specific synonymous shuffles H6, and exact 32-pair reverse-complement mirror shuffles H7). Cross-species transfer is two-scale: a symmetric vertebrate-wide macro-layer ($d_{2b} > 0.02$ across 132/132 mammalian pairs and 20/20 four-class vertebrate pairs) and a fine, order-structured mammalian codon-choice core governed by a reproducible Primate–Rodent directional asymmetry, while extreme compositional contrasts are cleanly separated by a paired GC3/acidic-proteome guard. Finally, compartment-level audits across 14 human RefSeqGene loci (`v14.6`, led by the *p*53 family *TP53*, *TP63*, *TP73*, the *RAS* family *KRAS*, *NRAS*, *HRAS*, *BRCA1*, and *HBB*) show that across this benchmark panel, positive conditional syntax is confined to spliced exons ($d_{3b} > 0$ and higher Mirror_{L_1} in 14/14 loci, $p = 2^{-14} = 6.1 \times 10^{-5}$) whereas excised introns and unspliced pre-mRNAs exhibit negative residuals ($d_{3b} < 0$ across all three reading frames in 14/14 loci), supporting the operational transition from intronic *form* to spliced exonic *program*. Finally, our `v14.7`–`v14.8` forensic audits demonstrate that this 29-transition ($L = 30$ codon) program persists across giant DNA viruses and deep subsurface extremophiles, blindly identifies foreign transgenic insertions in synthetic cells (*JCVI-syn1.0*) and host-mimicking "Trojan" effectors (such as HPV-16 *E7* and Adenovirus *E3 RID-α*) in human viruses, tracks the endosymbiotic and retroviral assimilation clock, and proves that standard industrial `Max-CAI` codon optimization

collapses conditional codon-pair syntax ($d_{2b} < 0$) across 14/14 human benchmark genes.

Data and Code Availability

All RefSeq coding-sequence archives (`*_cds_from_genomic.fna.gz`) are publicly available from the NCBI RefSeq FTP repository (`https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/`) under the accession numbers listed in Table 1 and Appendix Table 15, and all RefSeqGene records (`NG_*`) are available via NCBI Entrez E-utilities under the accessions in Table 8. The autonomous verification pipelines `super_codon_autonomous_v14.py` (`v14.5 audit-clean`) and `super_codon_proof_v14_6.py` (`v14.6`), including built-in startup unit tests, JSON and Markdown report generators, and exact command-line reproduction manifests, are documented in Appendix A and Appendix E.

A Reproducibility (`super_codon_autonomous_v14.5` and `v14.6`)

The complete primary audit, all seven extended biosphere audits, and the 14-gene RefSeqGene intron–exon (including *p*53 and *RAS* paralog families) audit are reproduced by `super_codon_autonomous_v14.py` (`v14.5 audit-clean`) and `super_codon_proof_v14_6.py` (`v14.6`). Note that the H4 bootstrap inner-shuffle count is fixed at `inner_reps=2` in `bootstrap_h4()`, whereas the H5 inner-surrogate count is exposed via `--h5-bootstrap-inner-reps` (default `2` in `full` mode).

```
# 1. Run built-in genetic code (tables 1,2,3,4,5,6,11,12), URL & splicing self-tests
#    (Stage 00 runs automatically before Stage 0 on every invocation):
python3 super_codon_autonomous_v14.py --out runs/v14_5_selftest \
    --cache cds_cache --offline --skip-smoke --skip-full
# (or standalone exit-0 unit tests):
python3 super_codon_autonomous_v14.py --self-test
python3 super_codon_proof_v14_6.py --self-test
# Verified output:
#   [self-test] Mirror pairs    : 32 pairs OK
#   [self-test] Splicing parser : 3-exon / 2-intron join(...) reconstruction OK
#   [self-test] ALL INVARIANTS PASSED.

# 2. Reproduce TP53+RAS exon/hotspot mutation audit, RAS family audit, and full 14-gene audit (v14.6):
python3 super_codon_proof_v14_6.py --intron-exon-genes TP53,KRAS,NRAS,HRAS \
    --cds-dir cds_cache --out-dir results_p53_ras_mutations --seed 20261011
python3 super_codon_proof_v14_6.py --ras-audit \
    --cds-dir cds_cache --out-dir results_ras_audit --seed 20261011
python3 super_codon_proof_v14_6.py --intron-exon-audit \
    --cds-dir cds_cache --seed 20261011
# 2b. Reproduce 13-genome Frontier Biosphere & Synthetic Genome Forensics Audit (v14.7):
python3 super_codon_proof_v14_6_7.py --frontier-audit \
    --cds-dir cds_cache --out-dir results_frontier_audit --seed 20261011
# 2c. Reproduce Viral Trojan, HGT Firewall, Organelle OS & Biotech Max-CAI Crash Audit (v14.8):
python3 super_codon_proof_v14_8.py --chimera-audit \
    --cds-dir cds_cache --out-dir results_chimera_v14_8 --seed 20261011
# (Note: v14.5 multi-species audits use seed=20261014; v14.6 uses default
#  seed=20261011, which affects only auxiliary Monte Carlo H5-H7 diagnostics.)

# 3. Download/verify all 55 RefSeq CDS archives (13 core + 1 virus + 41 extended):
python3 super_codon_autonomous_v14.py --out runs/v14_5_full_cache_download \
    --cache cds_cache --include-extended --include-virus \
    --skip-smoke --skip-full --download-timeout 1800 --seed 20261014

# 4. Reproduce primary 13-taxon + SARS-CoV-2 full audit (v14_5_paper_final):
python3 super_codon_autonomous_v14.py --out runs/v14_5_paper_final \
```

```
    --cache cds_cache --offline --include-virus --family-split \
    --seed 20261014 --skip-smoke --max-records 8000 --min-trans 100 \
    --reps 120 --h4-reps 8 --h4-bootstrap-reps 40 --h4-length 30 \
    --h4-threshold 0.0005 --h5-reps 3 --h5-bootstrap-reps 20 \
    --h5-bootstrap-inner-reps 2 --h5-threshold 0.001 --h6-reps 4 \
    --h6-bootstrap-reps 20 --h6-threshold 0.001 --h7-reps 4 \
    --h7-bootstrap-reps 20 --h7-threshold 0.001 --h5-transfer-reps 12 \
    --h5-conservation-max-gap 0.005 --h5-near-miss-max-gap 0.012 \
    --h5-conservation-min-ratio 0.80 --h5-conservation-max-ratio 1.25 \
    --gc-artifact-gc3-distance 0.12 --gc-artifact-acid-distance 0.08 \
    --virus-acc GCF_009858895 --virus-asm ASM985889v3

# 5. Reproduce 12-mammal radiation audit (v14_5_ext_mammals):
python3 super_codon_autonomous_v14.py --out runs/v14_5_ext_mammals \
    --cache cds_cache --offline --include-extended --family-split \
    --seed 20261014 --skip-smoke --max-records 8000 --min-trans 100 \
    --reps 120 --h4-reps 8 --h4-bootstrap-reps 40 --h4-length 30 \
    --h4-threshold 0.0005 --h5-reps 3 --h5-bootstrap-reps 20 \
    --h5-bootstrap-inner-reps 2 --h5-threshold 0.001 --h6-reps 4 \
    --h6-bootstrap-reps 20 --h6-threshold 0.001 --h7-reps 4 \
    --h7-bootstrap-reps 20 --h7-threshold 0.001 --h5-transfer-reps 4 \
    --h5-conservation-max-gap 0.005 --h5-near-miss-max-gap 0.012 \
    --h5-conservation-min-ratio 0.80 --h5-conservation-max-ratio 1.25 \
    --gc-artifact-gc3-distance 0.12 --gc-artifact-acid-distance 0.08 \
    --only Human_T2T Mouse Chimp Rattus_norvegicus Macaca_mulatta \
    Gorilla_gorilla Bos_taurus Sus_scrofa Canis_lupus_familiaris \
    Felis_catus Equus_caballus Oryctolagus_cuniculus

# 6-11. Reproduce the remaining six extended domain-specific audits
#       (using the identical full-audit flags as command 5 above,
#       replacing --out and --only as follows):
#
# 6. Four-class vertebrate bridge (runs/v14_5_ext_vert):
#    --out runs/v14_5_ext_vert --only Human_T2T Zebrafish Chicken \
#    Xenopus_tropicalis Takifugu_rubripes
#
# 7. Invertebrate audit (runs/v14_5_ext_inv):
#    --out runs/v14_5_ext_inv --only Drosophila C.elegans \
#    Anopheles_gambiae Apis_mellifera Bombyx_mori Caenorhabditis_briggsae
#
# 8. Land plants and green alga audit (runs/v14_5_ext_plants):
#    --out runs/v14_5_ext_plants --only Arabidopsis Oryza_sativa Zea_mays \
#    Solanum_lycopersicum Glycine_max Populus_trichocarpa \
#    Chlamydomonas_reinhardtii
#
# 9. Fungal clade audit (runs/v14_5_ext_fungi):
#    --out runs/v14_5_ext_fungi --only Yeast Schizosaccharomyces_pombe \
#    Candida_albicans Aspergillus_nidulans Neurospora_crassa
#
# 10. Protist clade audit (runs/v14_5_ext_protists):
#     --out runs/v14_5_ext_protists --only Plasmodium_falciparum \
#     Trypanosoma_brucei Leishmania_major Toxoplasma_gondii \
#     Dictyostelium_discoideum
#
# 11. Prokaryotic (bacteria, cyanobacteria, archaea) audit (runs/v14_5_ext_bact):
#     --out runs/v14_5_ext_bact --only E.coli B.subtilis M.jannaschii \
#     H.salinarum Pseudomonas_aeruginosa Staphylococcus_aureus \
#     Streptococcus_pyogenes Mycobacterium_tuberculosis Salmonella_enterica \
#     Lactococcus_lactis Helicobacter_pylori Bacteroides_fragilis \
#     Clostridioides_difficile Synechocystis_sp_PCC6803 Thermus_thermophilus
```

B Exact 95% bootstrap confidence intervals (`v14_5_paper_final`)

Species	d_{2b} 95% CI	d_3 95% CI	d_{3b} 95% CI	$\Delta_{H4}(30)$ 95% CI	Δ_5 95% CI	Δ_6 95% CI	Δ_7 95% CI
Drosophila	+0.0322..+0.0351	+0.1742..+0.1925	+0.1466..+0.1642	+0.00365..+0.00403	+0.0589..+0.0715	+0.0919..+0.0960	+0.0665..+0.0702
Mouse	+0.0524..+0.0554	+0.1256..+0.1313	+0.1055..+0.1107	+0.00468..+0.00508	+0.1118..+0.1174	+0.1278..+0.1311	+0.1103..+0.1135
Arabidopsis	+0.0240..+0.0255	+0.1233..+0.1282	+0.1007..+0.1048	+0.00241..+0.00275	+0.0486..+0.0529	+0.0696..+0.0726	+0.0535..+0.0552
Zebrafish	+0.0475..+0.0499	+0.1316..+0.1376	+0.1057..+0.1118	+0.00464..+0.00504	+0.0982..+0.1044	+0.1284..+0.1316	+0.1099..+0.1120
C. elegans	+0.0262..+0.0279	+0.1200..+0.1311	+0.0910..+0.0996	+0.00291..+0.00335	+0.0533..+0.0614	+0.0776..+0.0811	+0.0638..+0.0661
E. coli	+0.0426..+0.0467	+0.2633..+0.2785	+0.2093..+0.2210	+0.00526..+0.00580	+0.0844..+0.1035	+0.1513..+0.1567	+0.0979..+0.1021
M. jannaschii	+0.0607..+0.0662	+0.2606..+0.2729	+0.2181..+0.2301	+0.00590..+0.00713	+0.1290..+0.1410	+0.1801..+0.1871	+0.1317..+0.1399
B. subtilis	+0.0394..+0.0435	+0.1904..+0.2007	+0.1503..+0.1578	+0.00376..+0.00439	+0.0814..+0.0863	+0.1231..+0.1288	+0.0970..+0.1005
H. salinarum	+0.0341..+0.0378	+0.3464..+0.3681	+0.2878..+0.3055	+0.00444..+0.00515	+0.0736..+0.0960	+0.1011..+0.1067	+0.0780..+0.0830
Yeast	+0.0101..+0.0109	+0.1353..+0.1417	+0.1202..+0.1251	+0.00111..+0.00138	+0.0223..+0.0267	+0.0276..+0.0292	+0.0245..+0.0257
Chimp	+0.0470..+0.0506	+0.1371..+0.1454	+0.1158..+0.1237	+0.00449..+0.00483	+0.1010..+0.1085	+0.1242..+0.1281	+0.1076..+0.1106
Human_T2T	+0.0461..+0.0487	+0.1365..+0.1442	+0.1158..+0.1225	+0.00438..+0.00478	+0.0943..+0.1038	+0.1231..+0.1286	+0.1074..+0.1109
Chicken	+0.0435..+0.0461	+0.1384..+0.1445	+0.1158..+0.1221	+0.00467..+0.00505	+0.0865..+0.0945	+0.1268..+0.1294	+0.1116..+0.1144
SARS-CoV-2	−0.0678..−0.0207	−0.4291..−0.4173	−0.4326..−0.4161	−0.0236..+0.0029	−0.5005..−0.4108	−0.0472..+0.1062	+0.0104..+0.0773

Table 14: Exact 95% bootstrap confidence intervals (2.5th to 97.5th percentiles) for all seven strict null-model excesses in `v14_5_paper_final`. All 13 cellular species have strictly positive lower bounds across the entire stack.

C Complete within-species results for the 41 extended biosphere taxa

Species	Clade	RefSeq Accession	GC3	Acid	d_{2b}	d_3	d_{3b}	$\Delta_{H4}(30)$	Δ_5	Δ_6	Δ_7	Verdict
Extended Mammals (v14_5_ext_mammals):												
Rattus_norvegicus	mammal	GCF_015227675	0.5794	0.1153	+0.05413	+0.12540	+0.10514	+0.00490	+0.11431	+0.12872	+0.11076	STRICT
Macaca_mulatta	mammal	GCF_003339765	0.5828	0.1169	+0.04665	+0.13865	+0.11667	+0.00451	+0.09873	+0.12360	+0.10797	STRICT
Gorilla_gorilla	mammal	GCF_000151905	0.5739	0.1171	+0.04783	+0.13638	+0.11510	+0.00464	+0.09952	+0.12424	+0.10768	STRICT
Bos_taurus	mammal	GCF_002263795	0.6108	0.1165	+0.04225	+0.14755	+0.12556	+0.00437	+0.08264	+0.11554	+0.10090	STRICT
Sus_scrofa	mammal	GCF_000003025	0.6166	0.1143	+0.04315	+0.15096	+0.12907	+0.00442	+0.08720	+0.11683	+0.10371	STRICT
Canis_lupus_fam.	mammal	GCF_000002285	0.6001	0.1163	+0.04503	+0.15051	+0.12896	+0.00453	+0.08615	+0.12199	+0.10463	STRICT
Felis_catus	mammal	GCF_000181335	0.6072	0.1164	+0.04076	+0.14983	+0.12810	+0.00418	+0.08602	+0.10873	+0.09514	STRICT
Equus_caballus	mammal	GCF_002863925	0.6027	0.1160	+0.04494	+0.14360	+0.12303	+0.00448	+0.08484	+0.11943	+0.10456	STRICT
Oryctolagus_cunic.	mammal	GCF_000003625	0.6180	0.1135	+0.04084	+0.14358	+0.12045	+0.00417	+0.08142	+0.11209	+0.09915	STRICT
Extended Non-Mammalian Vertebrates (v14_5_ext_vert):												
Xenopus_tropicalis	vert.	GCF_000004195	0.4753	0.1192	+0.03965	+0.10930	+0.08927	+0.00372	+0.08593	+0.09802	+0.08330	STRICT
Takifugu_rubripes	vert.	GCF_901000725	0.6411	0.1205	+0.04242	+0.15577	+0.12879	+0.00433	+0.08974	+0.11713	+0.10621	STRICT
Extended Invertebrates (v14_5_ext_inv):												
Anopheles_gambiae	insect	GCF_000005575	0.6977	0.1152	+0.03872	+0.23709	+0.18771	+0.00502	+0.08089	+0.12130	+0.08866	STRICT
Apis_mellifera	insect	GCF_003254395	0.3303	0.1235	+0.01106	+0.16904	+0.15409	+0.00200	+0.01798	+0.04638	+0.03883	STRICT
Bombyx_mori	insect	GCF_014905235	0.4964	0.1209	+0.00787	+0.12286	+0.11047	+0.00157	+0.01650	+0.04081	+0.02854	STRICT
C. briggsae	nemat.	GCF_000004555	0.4311	0.1231	+0.02934	+0.13364	+0.10137	+0.00350	+0.06093	+0.08631	+0.07110	STRICT
Extended Plants & Green Algae (v14_5_ext_plants):												
Chlamydomonas_rein.	alga	GCF_000002595	0.8143	0.0910	+0.04923	+0.25795	+0.22223	+0.00737	+0.10462	+0.14407	+0.11366	STRICT
Populus_trichocarpa	plant	GCF_000002775	0.3967	0.1176	+0.03190	+0.10899	+0.09572	+0.00274	+0.06700	+0.07762	+0.06519	STRICT
Glycine_max	plant	GCF_000004515	0.4174	0.1175	+0.03074	+0.11484	+0.10104	+0.00306	+0.06363	+0.08238	+0.06957	STRICT
Zea_mays	plant	GCF_000005005	0.6090	0.1152	+0.01872	+0.17374	+0.15804	+0.00343	+0.03799	+0.09333	+0.06971	STRICT
Solanum_lycopersicum	plant	GCF_000188115	0.3637	0.1181	+0.02411	+0.11751	+0.10288	+0.00211	+0.05143	+0.06051	+0.05051	STRICT
Oryza_sativa	plant	GCF_001433935	0.6084	0.1154	+0.02226	+0.19733	+0.17956	+0.00464	+0.03994	+0.12760	+0.08782	STRICT
Extended Fungi (v14_5_ext_fungi):												
S. pombe	fungus	GCF_000002945	0.3279	0.1190	+0.00937	+0.14343	+0.12596	+0.00130	+0.01725	+0.02847	+0.02338	STRICT
Aspergillus_nidulans	fungus	GCF_000149205	0.5781	0.1179	+0.01245	+0.10957	+0.09266	+0.00144	+0.02439	+0.03480	+0.02981	STRICT
Neurospora_crassa	fungus	GCF_000182925	0.6463	0.1207	+0.01813	+0.15596	+0.12943	+0.00266	+0.03536	+0.06256	+0.05299	STRICT
Candida_albicans†	fungus	GCF_000182965	0.2759	0.1237	+0.02547	+0.19751	+0.17013	+0.00311	+0.05022	+0.07395	+0.06109	STRICT†
Extended Protists (v14_5_ext_protists):												
Trypanosoma_brucei	protist	GCF_000002445	0.5148	0.1208	+0.01718	+0.10506	+0.09106	+0.00246	+0.03106	+0.05156	+0.04586	STRICT
Leishmania_major	protist	GCF_000002725	0.7466	0.1087	+0.03418	+0.24347	+0.21018	+0.00382	+0.13386	+0.08979	+0.06339	SMALL_SAMPLE*
Plasmodium_falcip.‡	protist	GCF_000002765	0.1726	0.1365	+0.01073	+0.22733	+0.21222	+0.00283	+0.032580	+0.032575	+0.04093	STRICT
Dictyostelium_disc.	protist	GCF_000004695	0.1416	0.1112	+0.02771	+0.35057	+0.30555	+0.00576	+0.05295	+0.09288	+0.10067	STRICT
Toxoplasma_gondii	protist	GCF_000006565	0.6044	0.1244	+0.02400	+0.10436	+0.08255	+0.00428	+0.04384	+0.08832	+0.08829	STRICT
Extended Bacteria & Cyanobacteria (v14_5_ext_bact):												
Pseudomonas_aerug.	bact.	GCF_000006765	0.8803	0.1143	+0.06455	+0.46461	+0.36338	+0.00832	+0.14352	+0.19981	+0.13082	STRICT
Streptococcus_pyog.	bact.	GCF_000006785	0.3189	0.1215	+0.03160	+0.21525	+0.18188	+0.00307	+0.05606	+0.08993	+0.06580	STRICT
Lactococcus_lactis	bact.	GCF_000006865	0.2534	0.1222	+0.03083	+0.23247	+0.20852	+0.00292	+0.06402	+0.09898	+0.06453	STRICT
Salmonella_enterica	bact.	GCF_000006945	0.5968	0.1085	+0.05084	+0.26959	+0.21343	+0.00587	+0.10365	+0.15496	+0.09955	STRICT
Thermus_thermophilus	bact.	GCF_000008125	0.9177	0.1212	+0.12137	+0.53481	+0.41546	+0.01438	+0.31045	+0.38366	+0.28918	STRICT
Helicobacter_pylori	bact.	GCF_000008525	0.4189	0.1141	+0.09333	+0.33316	+0.27686	+0.01302	+0.21440	+0.37072	+0.28949	STRICT
Clostridioides_diff.	bact.	GCF_000009205	0.1733	0.1312	+0.01547	+0.22141	+0.20430	+0.00179	+0.01784	+0.05691	+0.03907	STRICT
Synechocystis_sp_PCC6803	cyano.	GCF_000009725	0.4903	0.1119	+0.06644	+0.24813	+0.22334	+0.00773	+0.13896	+0.21390	+0.17090	STRICT
Staphylococcus_aur.	bact.	GCF_000013425	0.2231	0.1242	+0.02251	+0.31205	+0.27214	+0.00178	+0.09069	+0.05316	+0.03139	SMALL_SAMPLE*
Bacteroides_fragilis	bact.	GCF_000025985	0.4544	0.1196	+0.03541	+0.20120	+0.16313	+0.00288	+0.07030	+0.10496	+0.08349	STRICT
Mycobacterium_tuberc.	bact.	GCF_000195955	0.7987	0.1051	+0.04577	+0.24710	+0.19196	+0.00449	+0.10174	+0.11430	+0.08834	STRICT

Table 15: Complete within-species log-likelihood excesses (in nats per codon transition) for all 41 extended RefSeq assemblies across the seven extended domain audits. All 41 assemblies have strictly positive point estimates and 95% bootstrap lower bounds across all seven nulls ($d_{2b}, d_3, d_{3b}, \Delta_{H4}(30), \Delta_5, \Delta_6, \Delta_7$). (*Two assemblies are flagged SMALL_SAMPLE solely due to $<$ 50,000 test transitions under locus_tag prefix grouping, while still maintaining strictly positive 95% bootstrap CIs across all seven quantities: *L. major* has d_{2b} CI = $[+0.0317, +0.0373]$, d_3 $[+0.2280, +0.2652]$, d_{3b} $[+0.1979, +0.2284]$, $\Delta_{H4}(30)$ $[+0.00336, +0.00477]$, Δ_5 $[+0.1147, +0.1443]$, Δ_6 $[+0.0847, +0.0928]$, and Δ_7 $[+0.0588, +0.0666]$; *S. aureus* has d_{2b} CI = $[+0.0183, +0.0275]$, d_3 $[+0.2869, +0.3296]$, d_{3b} $[+0.2523, +0.2868]$, $\Delta_{H4}(30)$ $[+0.00064, +0.00284]$, Δ_5 $[+0.0731, +0.1036]$, Δ_6 $[+0.0428, +0.0572]$, and Δ_7 $[+0.0281, +0.0353]$. †*Candida albicans* was processed under default Table 1 due to missing explicit [transl_table=12] tags in NCBI RefSeq nuclear FASTA headers; see Section 7.2. ‡In *Plasmodium falciparum*, $\Delta_5 = +0.032580$ [CI = $+0.031156.. + 0.037717$] and $\Delta_6 = +0.032575$ [CI = $+0.031826.. + 0.034267$] coincide to five decimal places after rounding but derive from distinct surrogate/shuffle controls.)

D Focal 34-pair evaluated H5-transfer matrix in v14_5_ext_mammals

Source (S)	Target (T)	$\Delta_{5T}^{\text{transfer}}$	$\Delta_{5T}^{\text{within}}$	$\text{gap}_{S\to T}$	$\text{ratio}_{S\to T}$	GC3 dist	Acid dist	Status (v14_5_ext_mammals)
Human_T2T	Chimp	+0.101557	+0.103030	−0.001473	0.986	0.001	0.001	H5-CONSERVED
Chimp	Human_T2T	+0.099553	+0.096853	+0.002699	1.028	0.001	0.001	H5-CONSERVED
Human_T2T	Gorilla_gorilla	+0.097916	+0.099523	−0.001607	0.984	0.004	0.000	H5-CONSERVED
Gorilla_gorilla	Human_T2T	+0.100425	+0.096853	+0.003571	1.037	0.004	0.000	H5-CONSERVED
Human_T2T	Macaca_mulatta	+0.097927	+0.098734	−0.000806	0.992	0.005	0.000	H5-CONSERVED
Macaca_mulatta	Human_T2T	+0.098401	+0.096853	+0.001548	1.016	0.005	0.000	H5-CONSERVED
Mouse	Rattus_norvegicus	+0.115591	+0.114307	+0.001283	1.011	0.008	0.001	H5-CONSERVED
Rattus_norvegicus	Mouse	+0.115964	+0.113427	+0.002536	1.022	0.008	0.001	H5-CONSERVED
Human_T2T	Mouse	+0.108491	+0.113427	−0.004937	0.956	0.007	0.001	H5-CONSERVED
Mouse	Human_T2T	+0.116465	+0.096853	+0.019612	1.202	0.007	0.001	not conserved
Chimp	Mouse	+0.110272	+0.113427	−0.003155	0.972	0.008	0.000	H5-CONSERVED
Mouse	Chimp	+0.120119	+0.103030	+0.017089	1.166	0.008	0.000	not conserved
Human_T2T	Rattus_norvegicus	+0.109524	+0.114307	−0.004783	0.958	0.001	0.002	H5-CONSERVED
Rattus_norvegicus	Human_T2T	+0.117571	+0.096853	+0.020717	1.214	0.001	0.002	not conserved
Chimp	Rattus_norvegicus	+0.110884	+0.114307	−0.003423	0.970	0.000	0.001	H5-CONSERVED
Rattus_norvegicus	Chimp	+0.122037	+0.103030	+0.019007	1.184	0.000	0.001	not conserved
Macaca_mulatta	Rattus_norvegicus	+0.109375	+0.114307	−0.004932	0.957	0.003	0.002	H5-CONSERVED
Rattus_norvegicus	Macaca_mulatta	+0.117122	+0.098734	+0.018389	1.186	0.003	0.002	not conserved
Bos_taurus	Sus_scrofa	+0.090072	+0.087200	+0.002873	1.033	0.006	0.002	H5-CONSERVED
Sus_scrofa	Bos_taurus	+0.083297	+0.082639	+0.000658	1.008	0.006	0.002	H5-CONSERVED
Canis_lupus_familiaris	Felis_catus	+0.081876	+0.086021	−0.004145	0.952	0.007	0.000	H5-CONSERVED
Felis_catus	Canis_lupus_familiaris	+0.093484	+0.086148	+0.007336	1.085	0.007	0.000	H5-NEAR-MISS
Felis_catus	Human_T2T	+0.096678	+0.096853	−0.000175	0.998	0.029	0.001	H5-CONSERVED
Human_T2T	Felis_catus	+0.091754	+0.086021	+0.005733	1.067	0.029	0.001	H5-NEAR-MISS
Equus_caballus	Human_T2T	+0.092657	+0.096853	−0.004196	0.957	0.024	0.001	H5-CONSERVED
Human_T2T	Equus_caballus	+0.093897	+0.084842	+0.009055	1.107	0.024	0.001	H5-NEAR-MISS
Canis_lupus_familiaris	Human_T2T	+0.091078	+0.096853	−0.005776	0.940	0.022	0.001	H5-NEAR-MISS
Human_T2T	Canis_lupus_familiaris	+0.097565	+0.086148	+0.011417	1.133	0.022	0.001	H5-NEAR-MISS
Bos_taurus	Human_T2T	+0.089786	+0.096853	−0.007068	0.927	0.033	0.001	H5-NEAR-MISS
Human_T2T	Bos_taurus	+0.097036	+0.082639	+0.014396	1.174	0.033	0.001	not conserved
Sus_scrofa	Human_T2T	+0.089069	+0.096853	−0.007784	0.920	0.038	0.003	H5-NEAR-MISS
Human_T2T	Sus_scrofa	+0.104048	+0.087200	+0.016848	1.193	0.038	0.003	not conserved
Oryctolagus_cuniculus	Human_T2T	+0.087396	+0.096853	−0.009458	0.902	0.040	0.004	H5-NEAR-MISS
Human_T2T	Oryctolagus_cuniculus	+0.101337	+0.081421	+0.019916	1.245	0.040	0.004	not conserved

Table 16: Focal evaluated H5-transfer conservation matrix (all 34 directed mammalian pairs evaluated in the focal H5-transfer protocol: 18 H5-CONSERVED, 8 H5-NEAR-MISS, and 8 not conserved) from the 12-mammal radiation audit (v14_5_ext_mammals).

E Focal H5-transfer inventory across all eight full audits

In super_codon_autonomous_v14.5, the focal H5-transfer list is constructed from DEFAULT_H5_TRANSFER_PAIRS (40 core pairs), OPTIN_H5_TRANSFER_PAIRS["SARS-CoV-2"] (8 viral pairs when --include-virus is active), and EXTENDED_H5_TRANSFER_PAIRS (104 extended pairs when --include-extended is active). When a domain-specific audit is executed with --only, focal pairs whose source or destination species is not loaded in that run are logged as skipped_missing_species, and pairs failing the directional sample-size guard ($\geq$ 20 units and $\geq$ 50,000 transitions in the source training split and destination test split) are logged as skipped_small_sample. Table 17 provides the exact accounting across all eight full audits.

Audit run	Total list	Missing-species skips	Small-sample skips	Evaluated	H5-CONSERVED	H5-NEAR-MISS	GC/aa artifact / not cons.
v14_5_paper_final	48	0	8	40	2	3	22 / 13
v14_5_ext_mammals	144	110	0	34	18	8	0 / 8
v14_5_ext_vert	144	128	0	16	1	4	0 / 11
v14_5_ext_inv	144	132	0	12	1	1	8 / 2
v14_5_ext_plants	144	126	0	18	1	3	6 / 8
v14_5_ext_fungi	144	130	0	14	0	0	6 / 8
v14_5_ext_protists	144	138	1	5	0	0	5 / 0
v14_5_ext_bact	144	110	2	32	0	0	21 / 11

Table 17: Complete accounting of pre-specified focal H5-transfer pairs across all eight full audits in v14.5 (code_mismatch skips = 0 in every run).

References

[1] Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27, 379–423 and 623–656.

[2] Lander, E. S., Linton, L. M., Birren, B., et al. (2001). Initial sequencing and analysis of the human genome. *Nature*, 409, 860–921.

[3] Pruitt, K. D., Brown, G. R., Hiatt, S. M., et al. (2014). RefSeq: an update on molecular reference sequence resources. *Nucleic Acids Research*, 42(D1), D971–D976.

[4] Perez, J.-C. (2009). *Codex Biogenesis — Les 13 codes de l'ADN: Un savant révèle le langage caché de l'ADN* (Préface de C. J. Calleman). Marco Pietteur Éditeur (Collection Résurgence), Embourg / Liège, Belgium, ISBN 978-2-87434-044-4. `https://www.editionsmarcopietteur.com/90-codex-biogenesis.html`

[5] Perez, J.-C. (2010). Codon populations in single-stranded whole human genome DNA are fractal and fine-tuned by the golden ratio 1.618. *Interdisciplinary Sciences: Computational Life Sciences*, 2, 1–13.

[6] Perez, J.-C. (2013). The "3 Genomic Numbers" discovery: how our genome single-stranded DNA sequence is "self-designed" as a numerical whole. *Applied Mathematics*, 4, 37–53.

[7] Perez, J.-C. (2015). Deciphering Hidden DNA Meta-Codes — The Great Unification & Master Code of Biology. *Journal of Glycomics & Lipidomics*, 5:131.

[8] Perez, J.-C. (2017). A research trip from artificial intelligence neural networks pioneering towards numerical genetics and numerical genomics (Codon mirror symmetry, *BRCA1* splicing $3n$ periodicity, and Dragon curve attractors). Keynote lecture at the *CAMMBIO Kick-Off Event*, Hochschule Mannheim, Germany, 27 November 2017. `https://www.cammbio.hs-mannheim.de/institute/kick-off-event.html`

[9] Perez, J.-C. (2018). Cancer, quantum computing and *TP53* tumor suppressor gene mutations prediction. *Novel Approaches in Cancer Study*, 1(2), NACS.000507, 29–37. `https://doi.org/10.31031/NACS.2018.01.000507`

[10] Perez, J.-C., & Montagnier, L. (2021). Six fractal codes of life from bioatoms atomic mass to chromosomes numerical standing waves: Three breakthroughs in astrobiology, cancers and artificial intelligence. *International Journal of Research — GRANTHAALAYAH*, 9(9), 133–191. `https://doi.org/10.29121/granthaalayah.v9.i9.2021.4191`

[11] Perez, J.-C. (2022). Epigenetics theoretical limits of synthetic genomes: The cases of artificials *Caulobacter* (*C. eth-2.0*), *Mycoplasma mycoides* (*JCVI-Syn 1.0*, *JCVI-Syn 3.0* and *JCVI_3A*), *E. coli* and Yeast chr XII. *Global Journal of Science Frontier Research: G Bio-Tech & Genetics*, 22(2), Version 1.0, 35–47.

[12] Perez, J.-C. (2026). The Perez Hourglass: Unifying the origins of the universal genetic code and opening pathways to next-generation quantum computing (Addendum: Quantum computing architectures based on the Perez Hourglass model). *International Journal of Biomedical Science and Research*, 2(3), 01–13.

[13] Perez, J.-C., & Chauprade, C. (2026). *The Music of Genes. Unlocking the Genetic Symphony: Can DNA Truly "Sing" Our Health Secrets? An Exploration of Jean-Claude Perez's Theories (1994–1995) & Modern Genomic Reality* (including *Ki-ras*, *N-ras*, and *Ha-ras* 570-base oncogene diversity and point-mutation dissonance). Zenodo. `https://doi.org/10.5281/zenodo.20641694`

[14] Petoukhov, S. V., & Svirin, V. I. (2013). Fractal genetic nets and symmetry principles in long nucleotide sequences. *Symmetrion*, 23(3–4), 303–322.

[15] Bernardi, G. (2007). The neoselectionist theory of genome evolution. *Proceedings of the National Academy of Sciences*, 104, 8385–8390.

[16] Gutman, G. A., & Hatfield, G. W. (1989). Nonrandom utilization of codon pairs in *Escherichia coli*. *Proceedings of the National Academy of Sciences*, 86(10), 3699–3703.

[17] Coleman, J. R., Papamichail, D., Skiena, S., Futcher, B., Wimmer, E., & Mueller, S. (2008). Virus attenuation by genome-scale changes in codon pair bias. *Science*, 320(5884), 1784–1787.

[18] Raoult, D., Audic, S., Robert, C., Abergel, C., Renesto, P., Ogata, H., La Scola, B., Suzan, M., & Claverie, J.-M. (2004). The 1.2-megabase genome sequence of Mimivirus. *Science*, 306(5700), 1344–1350.

[19] Gibson, D. G., Glass, J. I., Lartigue, C., Noskov, V. N., Chuang, R.-Y., Algire, M. A., Benders, G. A., Montague, M. G., Ma, L., Moodie, M. M., et al. (2010). Creation of a bacterial cell controlled by a chemically synthesized genome. *Science*, 329(5987), 52–56.

[20] Hutchison, C. A., III, Chuang, R.-Y., Noskov, V. N., Assad-Garcia, N., Deerinck, T. J., Ellisman, M. H., Gill, J., Kannan, K., Karas, B. J., Ma, L., et al. (2016). Design and synthesis of a minimal bacterial genome. *Science*, 351(6280), aad6253.

[21] Imachi, H., Nobu, M. K., Nakahara, N., Morono, Y., Ogawara, M., Takaki, Y., Takano, Y., Uematsu, K., Ikuta, T., Ito, M., et al. (2020). Isolation of an archaeon at the prokaryote–eukaryote interface. *Nature*, 577(7791), 519–525.